

UTM

UNIVERSITI TEKNOLOGI MALAYSIA

SCSR2213 NETWORK COMMUNICATIONS

Task#1: Project Setup

Group 4: Linux

SECTION: 5

Lecturer: Dr. Adnan Khalid

Name	Matric Number
Garrah THabit	A24CS4013
Mohammed Abdlerahim	A24CS4026
LEE TUNG ERN	A24CS0103
MOHAMMED ALSAKKAF	A23CS4026

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- First floor explanation

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INTRODUCTION

As members of Linux, we are excited to present our very first project for the Network Communication course. The objective of this project is to design a basic yet functional network for a newly proposed two-storey building within the Faculty of Computing. This facility will host several key spaces, including computer labs, a video conferencing room, and a student lounge, each with its own set of technical requirements.

Our main task is to develop a simple and efficient floor plan that addresses the networking needs of the building. We aim to ensure that the network layout is cost-efficient, easy to maintain, and adaptable for future expansion. Throughout the project, we will pay close attention to designing a solid network infrastructure that aligns with the faculty's operational goals while prioritizing security and performance.

The success of this project will largely depend on our ability to collaborate effectively as a team and to deliver a clear and functional design. By combining technical knowledge with good teamwork, we hope to produce a reliable network plan that fulfills the current demands and anticipates future needs.

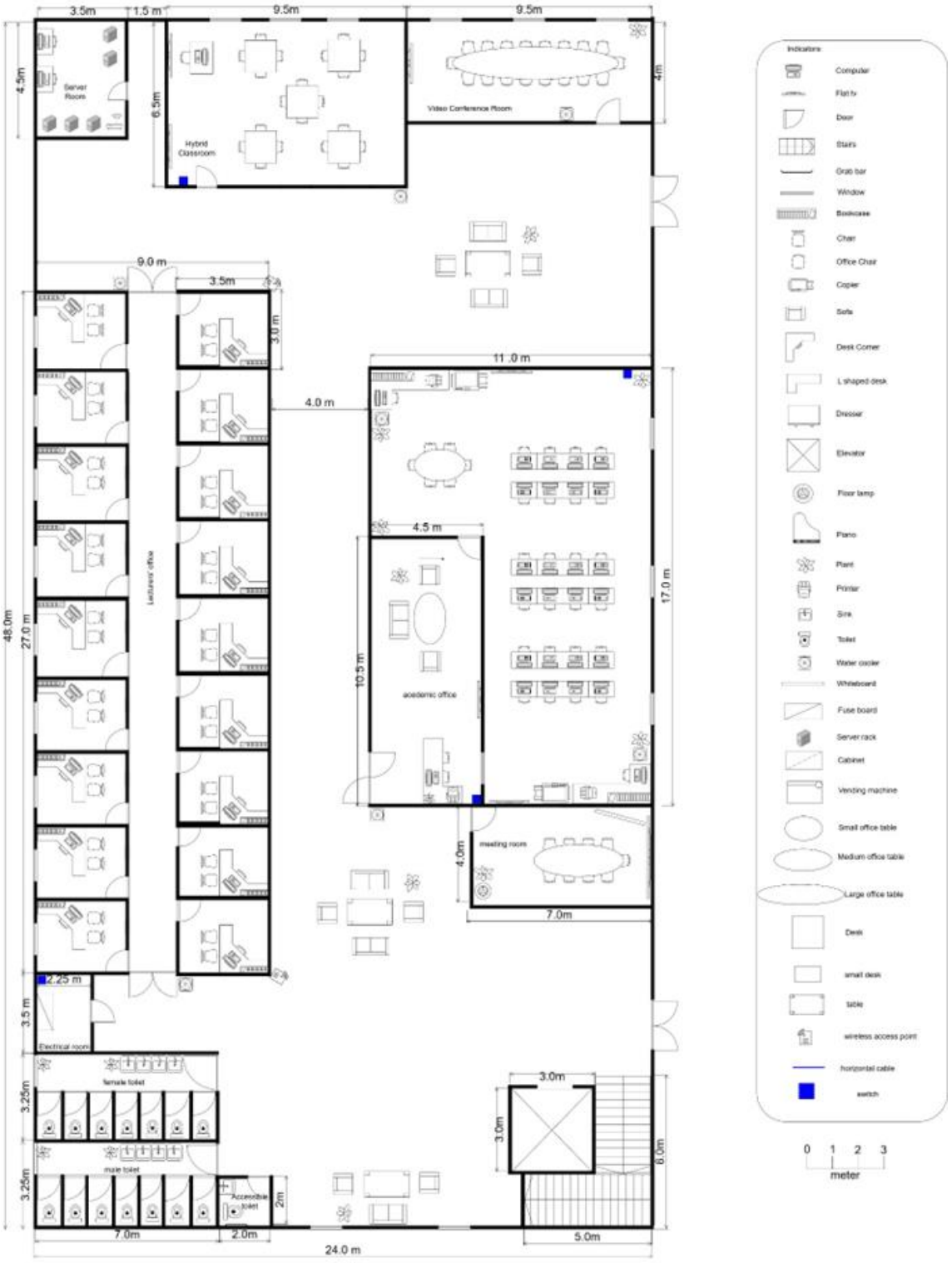
- **Explanation for the Floor Design:**

This is the newly designed Faculty of Computing building provides a modern and innovative learning space, built to accommodate a growing number of students and staff. It includes various essential facilities such as the academic office, faculty offices, server room, video conferencing room, four academic labs, a hybrid learning classroom, and a student lounge.

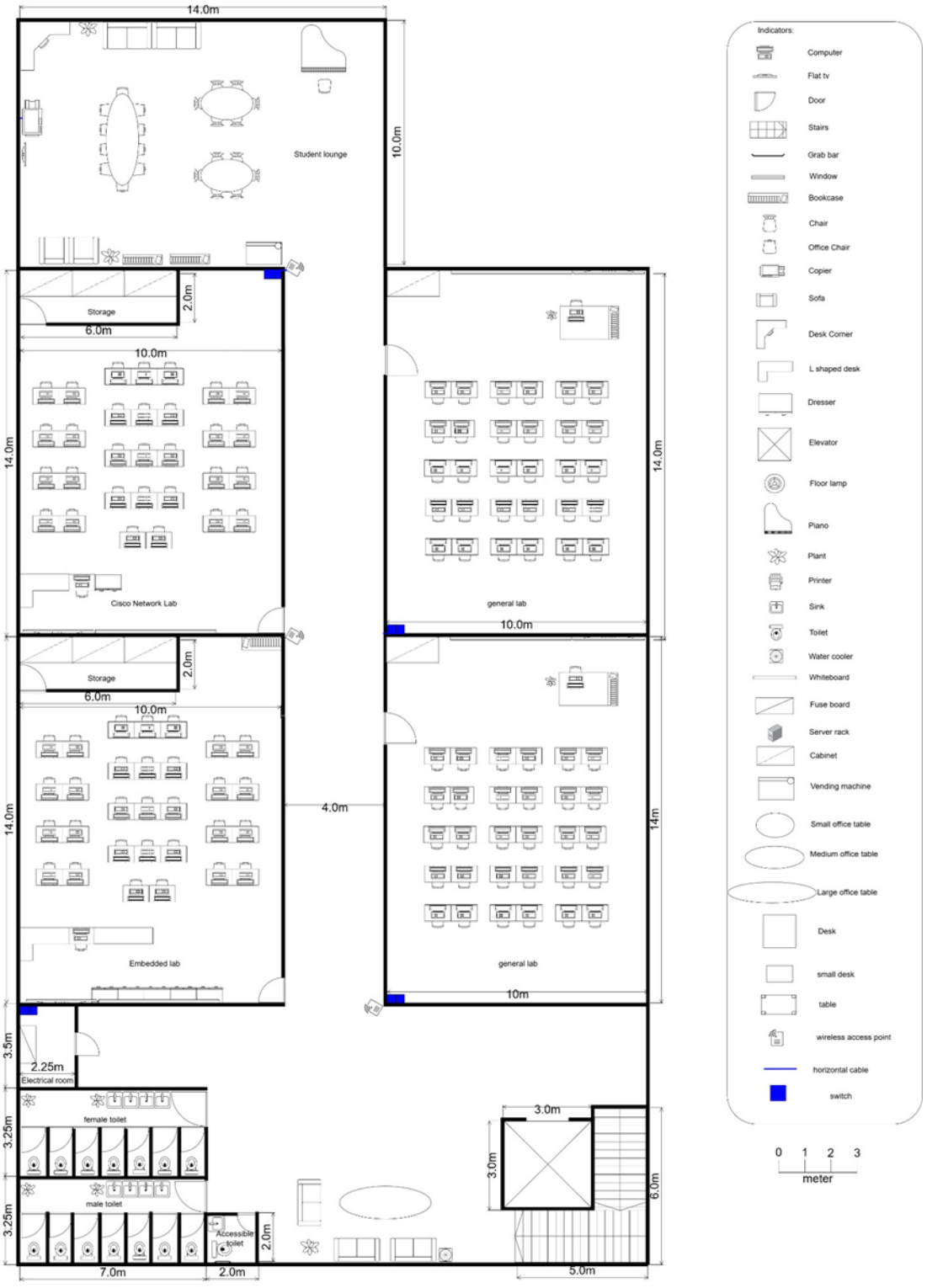
The **ground floor** is mainly designed to support the needs of faculty members and hybrid learning. From the upper entrance, one can immediately access the video conference room on the right side, followed by the hybrid learning space just a short distance further along the same side. Walking straight down the main hallway from this entrance, the server room can be found at the far end, accessible by taking a right turn. On the left side of this corridor, there is an entrance leading to the lecturers' offices. For those entering from the lower entrance, the staircase and elevator are located directly to the left. Continuing straight and turning right brings you to the academic office, while the toilets are situated further down the same corridor. Another convenient entry point to the faculty offices is also available nearby. Additionally, seating areas are placed near both entrances to provide comfort for staff and students while waiting.

The **first floor** focuses on academic laboratories and student facilities. Upon reaching the top of the stairs, the toilets are directly in front. Turning right and following the hallway leads to the lab areas. Two general-purpose labs are positioned on the right side, while the Cisco Network Lab and the Embedded Systems Lab are located on the left. These labs are equipped with specialized areas for technical equipment, and each lab contains 30 individual workstations. At the very end of the corridor is the student lounge, designed to be a relaxing area with seating, bookshelves, and even a piano for students to use. A copier machine is also available beside the lounge for students' convenience.

Ground Floor



First Floor


14.0m
Student lounge
10.0m
Storage
2.0m
6.0m
10.0m
14.0m
Cisco Network Lab
general lab
14.0m
10.0m
Storage
2.0m
6.0m
10.0m
14.0m
4.0m
Embedded lab
general lab
14m
10m
3.5m
2.25m
Electrical room
female toilet
3.25m
male toilet
3.25m
Accessible toilet
2.0m
7.0m
2.0m
3.0m
3.0m
6.0m
5.0m
Indicators:
Computer
Flat tv
Door
Stairs
Grab bar
Window
Bookcase
Chair
Office Chair
Copier
Sofa
Desk Corner
L shaped desk
Dresser
Elevator
Floor lamp
Piano
Plant
Printer
Sink
Toilet
Water cooler
Whiteboard
Fuse board
Server rack
Cabinet
Vending machine
Small office table
Medium office table
Large office table
Desk
small desk
table
wireless access point
horizontal cable
switch
0 1 2 3
meter

Meeting minutes

DATE	12 April 2025 1600H
LOCATION	N28
AGENDA	1. Discussion of project task 1 2. Task distribution for each member 3. Discussion of group name
MEETING MC	Lee Tung Ern

ATTENDANCE

Name	Time	Reason for absent
LEE TUNG ERN	1600H	-
GARRAH THABIT	1600H	-
MOHAMMED ABDELRAHIM	1600H	-
MOHAMMED ALSAKKAF	1600H	-

MINUTES

No	Item discuss	Idea / suggestion	Person in charge
1.	Group name	Group name decided to be “Linux”.	Mohammed Abdlerahim, Garrah Thabit
2.	Tool or software use to design the floor plan	Website “Draw io” decided to be used after discussion.	Garrah Thabit, Mohammed Abdlerahim, Lee Tung Ern
3.	Floor plan design	Tung Ern and Garrah designed the floor plan. Tung Ern volunteered to draw for the ground floor. Alsakkaf volunteered to draw for the first floor.	Lee Tung Ern, Mohammed Alsakkaf, Garrah Thabit
4.	Directory of floor	Tung Ern and Alsakkaf volunteered to complete the directory of floor.	Lee Tung Ern, Mohammed Alsakkaf
5.	Next meeting	14/14 - to gather everything	
	Meeting ended	6:00 pm	

Projected Marks

TASK 1		
ITEM	**MARKS**	**EXPECTED MARKS**
Group name and Members		
Suggested floor plan.		
Fit description – as per case study provided.		
A total 30 workstations and other equipment planned for each lab		
Scale: with scale		
Clear and readable. Has appropriate labels		

Creativity		
TOTAL		



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Faculty of Computing

SCSR2213 NETWORK COMMUNICATIONS

Task#2

Group 4: Linux
SECTION: 5
Lecturer: Dr. Adnan Khalid

Name	Matric Number
Garrah Thabit Mohammed	A24CS4013
Mohammed Abdulrahim	A24CS4026
LEE TUNG ERN	A24CS0103
MOHAMMED ALSAKKAF	A23CS4026

Primenlary Analysis:

Introduction

To understand the requirements of our networking lab project, we decided to prepare and answer a list of 11 questions. These questions were divided between general understanding and specific needs (hardware, software, bandwidth, and security). Our findings helped us determine the technical and operational feasibility of the proposed design and were used to support the overall planning.

☐

Questions and Answers:

1. **What devices are needed for the Cisco networking lab?**

A comprehensive Cisco networking lab requires several key devices. Switches are essential for connecting multiple devices within a local area network (LAN), facilitating efficient communication and resource sharing. Routers are necessary for connecting multiple networks and directing data traffic between them. Additionally, wireless access points (WAPs) enable wireless connectivity for mobile devices. The inclusion of firewalls, such as the Cisco ASA series, is critical to secure network traffic and support training in cybersecurity. For network simulation and design, software like Cisco Packet Tracer provides an invaluable platform for students to practice and refine their networking skills.

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2. **What bandwidth is required to ensure smooth operation for the video conferencing room?**

To ensure seamless video conferencing, adequate bandwidth must be provisioned. The recommended bandwidth per participant in group video calls is 8 Mbps, with an additional 20% overhead considered for protocol headers and encapsulation. Given that the conferencing room accommodates 18 users, the minimum base bandwidth required would be 144 Mbps. Including the overhead, the total recommended bandwidth increases to approximately 173 Mbps, ensuring uninterrupted, high-quality conferencing experiences.

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3. **What are the pros and cons of wireless connectivity for the student lounge?**

Wireless connectivity in student lounges offers several advantages. It provides flexibility, allowing students to access the network from anywhere within the lounge. It also facilitates collaborative learning, enabling real-time file sharing and group work. Wireless systems are generally easier and faster to install than wired alternatives and can accommodate multiple devices simultaneously. However, wireless networks may experience performance degradation during periods of high usage, and if not properly secured, they are vulnerable to unauthorized access and cyber threats.

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4. **Why is it important for labs to be equipped with technology that aligns with the 4th Industrial Revolution (4IR) standards?**

Integrating 4IR technologies into educational labs is crucial to align academic training with evolving industry demands. Exposure to advanced technologies such as artificial intelligence (AI), robotics, and the Internet of Things (IoT) enhances students' technical competencies and prepares them for the modern workforce. Furthermore, working with 4IR tools fosters the development of critical thinking, problem-solving, and effective communication skills, all of which are essential for success in today's dynamic professional environments.

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5. **What types of user authentication methods are generally recommended for educational networks?**

To secure sensitive academic and administrative data, educational institutions are advised to implement robust authentication mechanisms. Single Sign-On (SSO) simplifies access management by allowing users to authenticate once and gain access to multiple systems. Multi-Factor Authentication (MFA) adds an additional security layer by requiring secondary verification, such as a one-time code sent to the user's mobile device, thereby significantly enhancing protection against unauthorized access.

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6. **What physical and environmental requirements are recommended for server rooms in educational institutions?**

Server rooms should adhere to strict physical and environmental standards to ensure optimal equipment performance and longevity. The ambient temperature should be maintained between 18°C and 27°C, ideally around 22°C. The relative humidity should be controlled between 45% and 50% to prevent both moisture-related corrosion and electrostatic discharge. Implementing hot and cold aisle containment and utilizing HVAC systems equipped with real-time monitoring sensors are also recommended to maintain stable and efficient operating conditions.

7. **Are there common practices for implementing Quality of Service in educational networks to prioritize traffic for learning and administrative applications?**

Educational networks commonly deploy Quality of Service (QoS) techniques to prioritize time-sensitive applications such as video conferencing, VoIP, and e-learning platforms. Technologies such as Differentiated Services (DiffServ) and Resource Reservation Protocol (RSVP) are utilized to classify and manage network traffic. Additionally, network devices are configured with multiple queues to differentiate service levels. Centralized QoS policy management tools, like Cisco's QoS Policy Manager, enable IT teams to monitor and enforce QoS configurations campus-wide.

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8. **How is Wi-Fi coverage typically managed across faculty buildings to ensure reliable connectivity for students and staff in all areas**?

Achieving reliable Wi-Fi coverage across faculty buildings involves conducting comprehensive site surveys to identify high-density usage areas such as classrooms, lecture halls, and laboratories. High-capacity access points are strategically deployed to maximize coverage and performance. A centrally managed Wi-Fi system allows for streamlined control over network security and bandwidth allocation. Utilizing a Network-as-a-Service (NaaS) model can further enhance efficiency by outsourcing maintenance and scaling services according to institutional needs.

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9. **How can we improve the security of the desired system?**

Enhancing system security involves proactively addressing insider threats, ensuring all devices and communications are encrypted, and conducting regular security audits and penetration tests to identify and remediate vulnerabilities before they can be exploited.

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10. **What are the backup plans we should consider to prevent data loss?**

To effectively mitigate the risk of data loss, institutions should implement routine data backups, secure hardware physically, educate staff on cybersecurity best practices, maintain clean and organized work environments, deploy antivirus and anti-malware solutions, encrypt sensitive information, apply software patches promptly, and enforce strict security and password policies. Additionally, Data Loss Prevention (DLP) tools should be utilized to monitor and protect sensitive data across networks and endpoints.

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11. **What are the steps needed to acquire the utmost efficiency of the system?**

Maximizing system efficiency requires a multi-faceted approach, including upgrading network infrastructure, optimizing system configurations, efficient traffic management, and implementing proactive monitoring and maintenance strategies to detect and resolve issues before they impact system performance.

*** (note: the answers provided above is supported by references provided in the appendix below)**

Feasibility Analysis

This report summarizes the feasibility of implementing our Cisco networking lab project across three aspects:

A. Technical Feasibility

- The hardware and software required are readily available and well-supported by our institution's IT team.
- The technical complexity is manageable and aligned with the skills of the target users and support team.

B. Operational Feasibility

- The lab supports 4IR technology and serves undergraduate and postgraduate students.
- It enhances learning through exposure to realistic and advanced networking environments.

C. Economic Feasibility

- While initial costs may be high, long-term benefits (skill development, hybrid learning support) justify the investment.
- Bandwidth and hardware requirements are reasonable given the lab's expected output.

Conclusion: The Cisco lab, conferencing room, and student lounge Wi-Fi are all feasible from technical, operational, and economic standpoints. This investment aligns with the university's vision to prepare students for mode

- **This is a list of resources used for this project:**

1) NetworkLessons.com. (n.d.). Recommended lab equipment for Cisco CCNA. https://networklessons.com/cisco/ccna-routing-switching-icnd1-100-105/recommended-lab-equipment-for-cisco-c cna "reference for question 1"

2) NetDevGroup. (n.d.). NETLAB+ Requirements for Cisco Networking Academies. https://www.netdevgroup.com/content/cnap/requirements/ "reference for question 2"

3) Thomas Conner. (2024, March 3). Video conferencing bandwidth requirements. Targeting Mantra. https://targetingmantra.com/video-conferencing-bandwidth-requirements/ "reference for question 3"

4) Howard. (2024, July 30). Transforming education: Unlocking the benefits of Wi-Fi for schools. FS Community. https://community.fs.com/article/transforming-education-unlocking-the-benefits-of-wifi-for-schools.html "reference for question 4"

5) Huzaifa Khan. (2023, June 7). Preparing students for the Fourth Industrial Revolution: The role of 4IR skills in education. LinkedIn. https://www.linkedin.com/pulse/preparing-students-fourth-industrial-revolution-role-4ir-huzaifa-khan "reference for question 5"

6) Williams, J. J. (2021, November 10). *The Four Steps to Better User Authentication for Educational Institutions.* *James John Williams*. https://digt.com/informationsecurity/tpost/drxyme2je1-the-four-steps-to-better-user-authentica "reference for question 6"

7) *Temperature & Humidity: Standards for Data Centre & Server Rooms - UK Data Centre Cleaning*. (2024, October 11). UK Data Centre Cleaning. https://www.ukdatacentercleaning.co.uk/temperature-and-humidity/ "reference for question 7"

8) Fritz, J. (2020, February 14). QoS on Campus. *Technology Solutions That Drive Education*. https://edtechmagazine.com/higher/article/2008/02/qos-campus "reference for question 8"

9) Spectra. (2024, October 1). *Campus-Wide Managed Wi-Fi: From Lecture Halls to Hostels - Spectra*. https://www.spectra.co/resources/blog/from-lecture-halls-to-hostels-ensuring-seamless-managed-wi-fi-o n-campus "reference for question 9"

10) *Nile*. (2024, August 6). *Wi-Fi for University and College Campuses: How to Set It Up - Nile.* https://nilesecure.com/network-as-a-service/wi-fi-for-university-and-college-campuses-how-to-set-it-up "reference for questions 10, 11"

Meeting minutes

DATE	20 April 2025 1600H
LOCATION	Online ((google meet)
AGENDA	Discussion for task 2
MEETING MC	GARRAH THABIT

ATTENDANCE

Name	Time	Reason for absent
LEE TUNG ERN	9:00 PM	-
GARRAH THABIT	9:00 PM	-
MOHAMMED ABDELRAHIM	9:00 PM	-
MOHAMMED ALSAKKAF	9:00 PM	-

MINUTES

Name	Task
GARRAH THABIT	✓ Conducting research to find answers for the assigned questions. ✓ Compiling a list of references and sources. ✓ Designing questions to ask the user.
LEE TUNG ERN	✓ Writing a preliminary feasibility report.
MOHAMMED ALSAKKAF	✓ Creating a table aligned with the evaluation rubrics.
MOHAMMED ABDELRAHIM	✓ Drafting a report to document the meeting discussions. ✓ Developing a table that outlines the task requirements.

Projected Marks

TASK 1		
ITEM	**MARKS**	**EXPECTED MARKS**
Questions are appropriate to project and beneficial for better understanding		2
Questions are answered correctly and appropriately		2
Questions are researched through FC representative and reputable sources.		1
Answers are correctly referenced (with reference and citation)		2
Feasibility answer		1
Feasibility reasoning is logical and appropriate.		2
TOTAL		10

UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

SCSR2213 NETWORK COMMUNICATIONS

Task#3

CHOOSING THE APPROPRIATE LAN DEVICES

Group 4: Linux

SECTION: 5

Lecturer: Dr. Adnan Khalid

Name	Matric Number
Garrah Thabit Mohammed	A24CS4013
Mohammed Abdulrahim	A24CS4026
LEE TUNG ERN	A24CS0103
MOHAMMED ALSAKKAF	A23CS4026

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Introduction

In the previous tasks, we laid a solid foundation for our project. In **Task 1**, we formed our team, "Linux", and designed a detailed floor plan for the new two-story Faculty of Computing building. We also documented our initial discussions in clear meeting minutes.

In **Task 2**, we conducted a preliminary analysis by crafting over 10 targeted questions to clarify the requirements and gather insights. Through research and consultations, we identified the current and future needs of the project, evaluated its feasibility, and updated our design plan where necessary to align with stakeholder expectations.

Now, with these two steps completed, **Task 3** focuses on selecting the appropriate LAN devices with careful consideration of the given budget. This step involves identifying and justifying the network equipment that will meet the objectives of the case study. It is crucial to ensure that the new building is equipped with an efficient and secure network infrastructure to support both current and future demands.

After much research and considering the available resources, we agreed on selecting three brands for each of the network devices needed. The chosen devices are as follows:

1. **Printers**:
   A printer is a peripheral device that produces a durable representation of text or graphics, typically on paper. Various types of printers include 3D printers, inkjet printers, laser printers, and thermal printers.

2. **Copier**:
   A copier is a machine that makes copies of documents and other visual images onto paper or plastic film quickly and affordably. It is widely used in business, education, and government sectors.

3. **Smart TV**:
   A Smart TV is a traditional television set integrated with Internet connectivity and interactive Web 2.0 features, allowing users to stream music and videos, browse the web, and view photos. These devices combine computers, televisions, and digital media players, offering access to over-the-top media services like streaming TV and internet radio, along with network access.

4. **Routers**:
   A router is a network device that forwards data packets between computer networks. It connects one or more packet-switched networks or subnetworks, managing traffic between them and enabling multiple devices to share an Internet connection.

5. **Switches**:
   A switch is a network device that segments a local area network (LAN) into smaller segments. It improves network efficiency by directing traffic only to the intended device based on its MAC address. Switches support unicast, multicast, and broadcast communication within a single LAN segment.

6. **Desktop Tower**:
   A desktop tower is the main unit of a desktop computer, housing essential components such as the CPU, GPU, motherboard, RAM, storage, and power supply in a vertical case. It does not include peripherals like the monitor, keyboard, or mouse.

7. **Monitor**:
   A computer monitor is an output device that displays information in either pictorial or textual form. A monitor includes a visual display, supporting electronics, a power supply, housing, electrical connectors, and external user controls.

8. **Server**:
   A server is a device that provides functionality for clients, which are other programs or devices. Servers offer various services such as sharing data or resources among multiple clients or performing computations for clients.

9. **Cables**:
   Communication cables are used to interconnect devices like computers, routers, switches, and storage area networks. These cables are the medium through which data flows across the network.

10. **Fiber Optic Cables**:
    Fiber optic cables are high-speed communication cables that transmit data as light signals through thin strands of glass or plastic. They provide faster data transfer, greater bandwidth, and resistance to electromagnetic interference, making them ideal for long-distance and high-performance networking applications.

11. **Firewalls**:
    A hardware firewall is a network security solution designed to protect devices from malicious software and hardware attacks. It ensures that all traffic crosses the firewall for inspection, applying access controls and security policies to block unwanted traffic.

12. **Modems**:
    A modem is a device that connects a computer or network to the internet by converting digital signals into analog signals. It acts as a bridge between devices and an Internet Service Provider (ISP), enabling communication over telephone lines or mobile networks.

13. **Wireless Access Points**:
    A wireless access point (WAP) is a device that allows laptops, smartphones, and tablets to connect to a wired network via Wi-Fi. It serves as a hub for transmitting and receiving wireless signals, expanding network coverage in larger spaces.

14. **Patch Panels**:
    A patch panel is a hardware device used to organize and manage multiple cable connections, such as those between computers, servers, and switches. It provides a structured way to route and reroute network connections without handling individual cables directly, improving network organization and simplifying troubleshooting.

Table of Devices

Devices	Description	Price per unit (RM)	quantity	total price (RM)
1. Printer  **Brother A4 + A3 Ink Tank MFC-J2340DW**	**Fast Printing** With an impressive First Print Out Time (FPOT) of 4.4 seconds, 28 ipm(image per minute) for black and 28 ipm for colour, you spend less time waiting for prints, and more time crafting your ideas. **High Volume, Cost Efficient** Experience the versatile print formats with volumes at an incredibly cost efficient yield. With Brother's high-yield InkBenefit cartridges, get yields up to 3,000 pages for black and 1,500 pages for colour. Cost has never been more efficient. **Vivid Professional** Prints Great ideas materialise with quality prints. The new fast-drying pigment-based ink delivers superior print quality with high black density and true-to-life colours, ensuring each copy stays crisp, sharp and vivid.	1099	2	2198

2. Copier  **Brother MFC L9610CDN Color Laser All-in-One Copier**	**Fast business color printing** Provides the fastest color laser print speeds ever from Brother – up to 42 ppm(page per minute) to help increase workgroup productivity. **Create professional business documents.** Features the boldest laser color output ever from Brother with vivid color and sharp black output for the professional results your reports, brochures, presentations, and other important business documents require. **Lifetime value**. Offers built-in value with included Brother Genuine 6,500-page black and 5,000-page color toner cartridges. For even lower cost printing, use Brother Genuine TN810XL high-yield replacement toner cartridges for 12,000 pages black and 9,000 pages color. **Fast, high-volume scanning with**	6594	3	19782

	advanced capabilities. The 100-page capacity auto document feeder boasts duplex scan speeds up to 104ipm. Scan to popular destinations directly from the touchscreen display including cloud services, email, SharePoint, SSH Server (SFTP), and more. Create searchable and editable Microsoft Office documents including Word, Excel, and PowerPoint. Offers advanced scanning features such as scan preview, auto deskew, continuous scanning, blank page skip, and more. **Large, scalable paper handling.** Spend less time refilling paper trays with the 520-sheet capacity paper tray and 100-sheet capacity multipurpose tray. Total capacity is expandable up to 1,620 sheets with optional trays. An optional cabinet can be used to store paper and supplies and turn the printer into a floor-standing configuration. **Flexible connectivity.** Features built-in Gigabit Ethernet networking to seamlessly set up and share on your wired network, with the availability of an optional dual band (2.4GHz / 5GHz) wireless module (NC-9000W). Easily print from and scan to a wide variety of mobile devices.			
3. Smart TV  **PRISM+ Q75 Ultra**	The Q-Ultra Series, powered by Dolby Vision, HDR10+ and Filmmaker mode, makes for a cinematic experience like none other. Enjoy exceptional colors, contrast and lifelike details across the spectrum. Experience a total sound immersion with Dolby Atmos, a 3D spatial sound technology.	4099	12	49188

	4K UHD Definition Expect ultra-crisp and sharp visuals down to every single pixel. See every detail in its most accurate and life-like form. **Far Field Voice Control** Control your TV, hands-free using just your voice, and nothing else. With in-built microphones, command it using Google Assistant by simply saying "Hey Google or Ok Google" – a true wireless experience. **Seamless Screencasting** Seamlessly cast photos, videos and music from your phone to the QLED screen for an even greater immersive experience without suffering from resolution loss. **Space-Optimizing Solutions** Save on space with a swivel or fixed mounts that bring added flexibility to your setup.			
4. Routers  **TP-Link Archer AX75**	Super-Fast Tri-Band WiFi Up to 5.4 Gbps 2402 Mbps (5 GHz) + 2402 Mbps (5 GHz) + 574 Mbps (2.4 GHz) Connect More, Stay Fast - OFDMA and MU-MIMO deliver data to multiple devices simultaneously, expanding the available bandwidth Broad Coverage - 6× high-performance antennas, Beamforming, and high-power FEM combine to ensure vast coverage Powerful Quad-Core Processing - Equipped with a 1.7 GHz Quad-Core CPU to handle huge throughput and eradicate latency	970	4	3880

5. Switches	250 W PoE Budget: 24× 802.3at/af-compliant PoE+ ports with a total	1799	8	14392

 **TP-Link TL-SG2428P**	power supply of 250 W*. Full Gigabit Ports: 24× gigabit PoE+ ports and 4× gigabit SFP Slots provide high-speed connections. Integrated into Omada SDN: Zero-Touch Provisioning (ZTP)**, Centralized Cloud Management, and Intelligent Monitoring. Centralized Management: Cloud access and Omada app for ultra convenience and easy management. Static Routing: Helps route internal traffic for more efficient use of network resources. IPv6: IPv6 support with dual IPv4/IPv6 stack, MLD snooping, IPv6 neighbor discovery.			
6. Desktop tower  **Acer Consumer Desktop | Aspire XC Series XC-1785**	Operating System: Windows 11 Home Processor: Intel® Core™ i5-14400 Graphics: Intel® UHD Graphics Memory: 16 GB DDR5 (upgradable to 32 GB) Storage: 1 TB PCIe Gen4 SSD Ports: Front: Card reader, 2 USB 3.2 Gen 1 Type A, 1 USB 3.2 Gen 1 Type C, 2 audio jacks Rear: 4 USB 2.0 Type A, 1 USB 3.2 Gen 1 Type A, 2 HDMI, LAN port, 3 audio jacks Wireless: Wi-Fi 6	3299	172	567428

7. Monitor  **Acer Monitor - Mainstream Series - Work & Play| EK221Q H**	Resolution: 1920 x 1080 (Full HD) Refresh Rate: 100Hz Response Time: 1ms (VRB) Colour Gamut: 99% sRGB Panel Type: VA (Vertical Alignment) Brightness: 250 cd/m² Contrast Ratio: 3000:1 AMD FreeSync: Yes (HDMI VRR) Speakers: Yes (2 x 2W)	289	172	49708
8. Server  **HPE ProLiant DL20 Gen11 Rack Server (E-2434.16GB.1TB)**	Processor(s): Intel® Xeon® E-2434 (4-Core, 3.4 GHz, 55W) Memory: 16 GB UDIMM DDR5 4800 MT/s (1x 16 GB) Hard Disk: HPE 1TB SATA 7.2K LFF LP HDD Storage Controller: Embedded Intel® VROC SATA software RAID Network Controller: HPE embedded 1Gb 4-port BCM5719 network adapter Optical: HPE 9.5mm SATA DVD-ROM Optical Drive Power Supply: (1) 290W Non-redundant Power Supply	9883	1	9883

9. Ethernet Cables  **cat 6a (2996AS-5G) by Videk**	Patch Lead RJ45 Plug to RJ45 Plug SFTP Double shielded for protection against EM and RF interference Low crosstalk 10Gbps CAT-6a standard for high demand applications UL94V-2 Plug housing flammability	70.1/5 m	400	28040
10. Fiber Optic Cables	Fiber Optic Cable	652.88/ 100m	2	1305.76

 **Fiber Optic (MP007797) by MULTICOMP PRO**	OS2 Singlemode 8 Fibres 6 mm 328 ft 100 m			

11. Firewalls  **1U Rack Mount Firewall Appliance**	Intel Quad-Core processor for high-performance routing and firewall tasks. Up to 6 Gigabit Ethernet ports for flexible and scalable networks. Supports up to 8 GB DDR4 RAM for smooth handling of resource-heavy tasks like IDS/IPS and VPNs. Offers storage options (SSD or HDD) for logs, backups, and configurations. Compact 1U rack-mountable design, ideal for data centers and enterprise setups. Advanced cooling system for reliable performance under heavy workloads.	1200	2	2400
12. modems  **Nighthawk®Mid/high-Split Multi-Gig 2.5Gbps Cable Modem**	Supports very fast 2.5Gbps internet speeds for quick connectivity. Works with major internet providers like Xfinity, Spectrum, and Cox. Uses DOCSIS 3.1 technology for reliable high-speed performance. Has a multi-gig Ethernet port for easy connection to high-speed routers. Compact design with good cooling for steady operation. Great for future needs, supporting activities like 4K streaming.	1300	2	2600

13. wireless access point  **TP-Link EAP653 AX3000**	Supports Wi-Fi 6 (802.11ax) technology with speeds up to 2976 Mbps (2402 Mbps on 5 GHz and 574 Mbps on 2.4 GHz). Features MU-MIMO and OFDMA for efficient multi-device handling, reducing latency and improving network performance. Provides seamless roaming for uninterrupted connectivity while moving across the network. Includes Power over Ethernet (PoE+) support for easy installation and reduced cable clutter. Offers advanced security protocols, including WPA3, to protect user data and enhance network security. Compact and sleek design for ceiling or wall mounting, suitable for offices, schools, and hospitality environments.	580	14	8120
14. patch panel  **ICC Patch Panel, CAT 6A, 48-Port, 2 RMS (ICMPP0486B)**	Has **48 RJ45 ports** for connecting many devices with fast Cat6A Ethernet. Supports **10 Gigabit Ethernet** for reliable and quick data transfer. Designed as a **2U rack-mount**, fitting easily into standard 19-inch racks for a neat setup. Comes with **color-coded labels** for easy wiring, making setup and maintenance simpler. Made with a **strong metal frame** for durability in tough environments. Ideal for **large cabling projects** in places like data centers and server rooms.	1700	5	8500

Total price	767424.76

Reflections:

1- Are you surprised by the prices? How were you surprised?

Yes, we were quite surprised by the price of RM 767,424.76, which constitutes a significant portion of our budget. We didn't expect network devices and equipment to be so costly, especially since many of the items were not top models. Even basic items like cables added more to the overall cost than we anticipated. The total price was much higher than we had expected, showing us how quickly costs can accumulate.

2- Have you ever considered cost as a factor for choosing networking devices?

Cost was not our primary concern when selecting networking devices. Our focus was on finding devices that met the needs of our project and ensuring the network would function effectively within our budget. We evaluated various options and chose the ones that provided the best balance of performance, value, and suitability for our project.

3- What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

1.Printer

		
HP OfficeJet Pro 8120 All-in-One Printer	Canon Pixma E410 All-In-One Printer	Brother A4 + A3 Ink Tank MFC-J2340DW

Speed: ● Black: Up to 20 ppm (ISO) ● Color: Up to 10 ppm (ISO) ● Draft Mode: Up to 29 ppm (black), 25 ppm (color) Duplex Printing: Automatic (standard) Wireless Capability: Yes, with dual-band Wi-Fi (802.11 a/b/g/n/ac) Monthly Page Volume: Up to 800 pages Cartridges: 4 (Black, Cyan, Magenta, Yellow) Scan Resolution: Up to 1200 x 1200 dpi (optical) Additional Features: Touchscreen, Quiet mode, Mobile printing via HP Smart, AirPrint, Mopria, etc.	Print Speed: ● Mono: 8 ipm ● Color: 4 ipm Duplex Printing: Manual Wireless Capability: Yes (via USB 2.0) Monthly Page Volume: 200 Cartridges: 2 (PG-47 Black, CL-57S Color) with an optional CL-57 cartridge Scan Resolution: Up to 600 x 1200 dpi (optical) Additional Features: Quiet Mode, Mobile printing support (AirPrint, Canon print app), 48-bit color depth for scanning	Print Speed: ● Mono/Color: 28 ipm ● First Print Out Time (FPOT): 4.4 seconds (Mono), 4.6 seconds (Color) Duplex Printing: Automatic Wireless Capability: Yes (IEEE 802.11b/g/n) Monthly Page Volume: 1200 pages Cartridges: 1 (Black) and 1 (Color) Scan Resolution: Up to 600 x 600 dpi (optical) Additional Features: Color copying, high print resolution (up to 1200 x 4800 dpi), ADF (up to 50 sheets)
RM 786	RM219	RM 1099

We have chosen the Brother A4 + A3 Ink Tank MFC-J2340DW primarily due to its high-speed printing capabilities, which make it an optimal choice for high-volume office environments. This printer is capable of handling a higher monthly page volume compared to other models, ensuring it can meet the demands of substantial paper handling. While the HP OfficeJet Pro 8120 All-in-One Printer offers superior print quality, the primary consideration for our office needs is print speed, making the Brother printer a more suitable option. In contrast, the Canon Pixma E410 All-In-One Printer is a more affordable choice but is not ideal for our requirements as it is unable to efficiently handle large printing volumes.

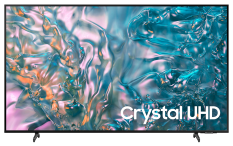
2. Copier

 HP LaserJet M635fht Laser Multifunction Printer-Monochrome-Copier	 Kyocera TASKalfa 2554ci Printer	 Brother MFC L9610CDN Color Laser All-in-One Copier
Type: Laser Multifunction Printer (Copier/Fax/Printer/Scanner) Print Speed: 65 ppm (monochrome) Print Resolution: 1200 x 1200 dpi Duplex Printing: Automatic Scanner: Flatbed, 600 dpi optical resolution, Color scanning Copier: Monochrome, Flatbed Fax: Monochrome, G3 Fax	Type: Color Multifunction Printer (Copier/Fax/Printer/Scanner) Print Speed: 25 ppm (color & monochrome) Print Resolution: 1200 x 1200 dpi (print), 600 x 600 dpi (scan) Duplex Printing: Standard Scanner: Flatbed and Document Feeder, 600 dpi resolution, Color scanning Copier: Color & Monochrome, Multiple copy features (Zoom, Electronic Sort, etc.) Fax: G3 Fax, 33.6 Kbps modem speed, 3-second transmission speed Media Handling: 2 x 500-sheet	Type: Color Laser Multifunction Printer (Copier/Fax/Printer/Scanner) Print Speed: 42 ppm (monochrome and color) Print Resolution: 2400 x 600 dpi Duplex Printing: Automatic Scanner: Flatbed/ADF, 600 x 600 dpi optical resolution, Color scanning Copier: Color, Flatbed Fax: Color, Flatbed
Media Handling: 1200-sheet capacity (combined input and output trays) Monthly Duty Cycle: 300,000 pages Recommended Monthly Print Volume: 5,000 to 30,000 pages Key Features: High print speed, large input capacity, and high monthly duty cycle. .	input trays, 150-sheet multipurpose tray, 100-sheet ADF Monthly Duty Cycle: 80,000 pages Recommended Monthly Print Volume: 10,000 to 25,000 pages Key Features: Compact design with high-quality color printing and advanced copy features	Media Handling: 520-sheet input tray, 100-sheet ADF, and 100-sheet multipurpose tray Monthly Duty Cycle: 120,000 pages Recommended Monthly Print Volume: 12,000 pages Key Features: High print quality and support for multiple wireless print technologies.

RM 18715	RM16880.46	RM 6594

We have selected the Brother MFC L9610CDN for its affordability relative to other alternatives, while still providing the essential features needed for faculty use. Its recommended monthly print volume of 12,000 pages is adequate to support the heavy printing demands within the faculty. While other options, such as the HP LaserJet M635fh and Kyocera TASKalfa 2554ci Printer, offer additional features and superior performance, the increased costs of these models—more than double the price of the Brother MFC L9610CDN—do not provide enough additional value to justify the higher price. As such, these alternatives are less budget-friendly for our needs..

3. Smart Tv

 PRISM+ Q75 Ultra	 Philips 7100 Series 75-inch 4K UHD LED TV	 SAMSUNG 75" Crystal UHD DUE800 4K Smart TV
Panel Technology: QLED Operating System: Google TV Screen Size: 75" Resolution: 4K (3840 x 2160) Refresh Rate: 60Hz HDR Support: HDR10+, Dolby Vision Dolby Atmos: Yes Voice Control: Hands-free	Panel Technology: 4K Ultra HD LED Operating System: Google TV Screen Size: 75" Resolution: 4K (3840 x 2160) Refresh Rate: Not specified HDR Support: HDR10, Dolby Vision, HLG Dolby Atmos: Yes Voice Control: Google Assistant	Panel Technology: LED Operating System: Tizen™ Screen Size: 75" Resolution: 4K (3840 x 2160) Refresh Rate: 50Hz HDR Support: HDR10+ Dolby Atmos: No Voice Control: Bixby, Alexa,
Google Assistant Connectivity Options: HDMI 2.0 (3), USB (2), Ethernet, Optical Audio Out Bluetooth Version: Bluetooth 5.1 Wall Mount VESA: 600 x 400 mm	with remote microphone Connectivity Options: HDMI (3), USB (2), Ethernet, Optical Audio Out Bluetooth Version: Bluetooth 5.1 Wall Mount VESA: 500 x 300 mm	and Google Assistant support Connectivity Options: HDMI (3), USB (2), Ethernet, Optical Audio Out Bluetooth Version: Bluetooth 5.2 Wall Mount VESA: 400 x 400 mm
RM 4099	RM 4399	RM 4899

We have chosen the PRISM+ Q75 Ultra for its superior features, including its QLED panel, which provides vibrant colors, enhanced brightness, and wider viewing angles compared to the Philips 7100 Series 75-inch 4K UHD LED TV and the SAMSUNG 75" Crystal UHD DUE800 4K Smart TV. Another significant factor in our decision was its ease of use; the hands-free Google Assistant and Google TV functionalities make it particularly suitable for faculty use. Additionally, the PRISM+ Q75 Ultra is lighter and offers greater wall mount compatibility, making installation and maintenance simpler. Lastly, it is the most cost-effective option among the three, providing excellent value for money.

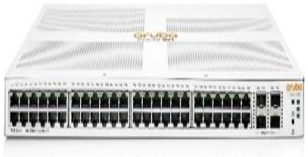
4-Routers:

		
ASUS RT-AX88U Pro	**Netgear RAX50**	**TP-Link Archer AX75**
• Wi-Fi 6 technology for faster and more reliable wireless connections • Dual-band network band for enhanced network performance	• UP TO 5.4GBPS SPEEDS—600+4800Mbps with 6-stream connectivity** • 160MHZ CHANNEL SUPPORT***—Enjoy greater than 1Gbps WiFi speeds for blazing fast	• **Super-Fast Tri-Band WiFi Up to 5.4 Gbps** - 2402 Mbps (5 GHz) + 2402 Mbps (5 GHz) + 574 Mbps (2.4 GHz) • **Connect More, Stay Fast** OFDMA and MU-MIMO

• Supports virtual assistant Alexa for voice control and convenience • Gigabit Ethernet technology for high-speed wired connections • USB port for easy file sharing and device connectivity • 5 network (RJ-45) ports and 1 broadband (RJ-45) port for versatile network setup • WPA3 and WPS encryption standards for secure wireless communication • Wi-Fi mesh system support for seamless coverage throughout your home • 4 external antennas for extended wireless range Total wireless transmission speed of 744 MB/s for smooth streaming and gaming	• streaming and downloading to newer mobile devices • SUPERSPEED USB 3.0 PORT —Up to 10x faster than USB 2.0 5 GIGABIT PORTS—Connect more wired devices for faster file transfer and uninterrupted connections with 1 WAN & 4 LAN Ethernet ports • 1024-QAM—25% increased data efficiency and faster speeds than a 256-QAM router	• deliver data to multiple devices simultaneously, expanding the available bandwidth • **Broad Coverage -** 6× high-performance antennas, Beamforming, and high-power FEM combine to ensure vast coverage **Powerful Quad-Core Processing -** Equipped with a 1.7 GHz Quad-Core CPU to handle huge throughput and eradicate latency
RM1,488	RM1,259.00	RM970

We have selected the TP-Link Archer AX75 due to its cost-effectiveness and its support for Wi-Fi 6 technology, offering speeds of up to 5.4Gbps. It is capable of managing multiple devices simultaneously through the use of MU-MIMO and OFDMA technologies. Additionally, its six antennas and tri-band support ensure reliable performance in high-traffic areas. The router will primarily serve for routing and redundancy purposes, with access points handling the majority of wireless connectivity needs.

5-Switches:

		 **Aruba Instant On 1930 48G**
TP-Link TL-SG2428P	**MikroTik CRS326-24G-2S+RM**	**4SFP/SFP Switch JL685A**
• 250 W PoE Budget: 24× 802.3at/af-compliant PoE+ ports with a total power supply of 250 W*. • Full Gigabit Ports: 24× gigabit PoE+ ports and 4× gigabit SFP Slots provide high-speed connections. • Integrated into Omada SDN: Zero-Touch Provisioning (ZTP)**, Centralized Cloud Management, and Intelligent Monitoring. • Centralized Management: Cloud access and Omada app for ultra convenience and easy management. • Static Routing: Helps route internal traffic for more efficient use of network resources. • IPv6: IPv6 support with dual IPv4/IPv6 stack, MLD snooping, IPv6 neighbor discovery.	• Non-blocking Layer 2 switching capacity • 16K host table • IEEE 802.1Q VLAN • Supports up to 4K VLANs • Port isolation • Port security • Broadcast storm control • Access Control List • Web-based GUI	• 48 Gigabit Ethernet ports and 4 SFP/SFP+ uplink ports for flexible and high-speed connectivity. • Throughput capacity (Mpps) : 130.95 Mpps • Switching capacity: 176 Gbps • Reliability MTBF (years): 114 • Maximum power rating: 36.9W
RM1,799.00	RM1,038.00	RM2,909.99

We have chosen the TP-Link TL-SG2428P due to its 24 Gigabit PoE+ ports, 4 SFP uplink ports, and a 250W PoE budget, which provides sufficient power for access points and other devices. Unlike other switches that do not support PoE, the TL-SG2428P operates effectively with access points without the

need for additional PoE injectors. While alternative models may offer more ports and higher switching capacity, their lack of a PoE budget and higher costs make them less practical for our needs.

6. Desktop

		
HP Pro Tower 400 G9 PCI Desktop PC	DELL OptiPlex Small Form Factor	Acer Consumer Desktop | Aspire XC Series - XC-1785
Operating System: Windows 11 Pro Processor: Intel® Core™ i5-14500 (up to 5.0 GHz, 24 MB L3 cache, 14 cores, 20 threads) Graphics: Intel® UHD Graphics 770 Memory: 8 GB RAM Storage: 512 GB SSD Ports: 1 headphone/microphone combo 1 SuperSpeed USB Type-C® (10Gbps) 3 SuperSpeed USB Type-A (10Gbps) Security: HP Wolf Security (hardware-enforced defense)	Operating System: Windows 11 Pro Processor: Intel® Core™ i5-14500 vPro® (up to 5.0 GHz, 24MB cache, 14 cores, 20 threads) Graphics: Intel® UHD Graphics Memory: 16 GB DDR5 RAM Storage: 512 GB PCIe NVMe SSD Ports: Front: 2 USB 2.0, 1 USB 3.2 Gen 1 (5 Gbps) Type-C®, 1 audio jack Rear: 2 USB 2.0, 2 USB 3.2 Gen 1, 1 DisplayPort™, 1 HDMI 1.4b, 1 RJ45 Ethernet Port Wireless: No Wi-Fi LAN card	Operating System: Windows 11 Home Processor: Intel® Core™ i5-14400 Graphics: Intel® UHD Graphics Memory: 16 GB DDR5 (upgradable to 32 GB) Storage: 1 TB PCIe Gen4 SSD Ports: Front: Card reader, 2 USB 3.2 Gen 1 Type A, 1 USB 3.2 Gen 1 Type C, 2 audio jacks Rear: 4 USB 2.0 Type A, 1 USB 3.2 Gen 1 Type A, 2 HDMI, LAN port, 3 audio jacks Wireless: Wi-Fi 6
RM 3637	RM 3661.87	RM 3299

We have selected the Acer Consumer Desktop | Aspire XC Series - XC-1785 primarily for its large storage capacity of 1TB SSD, which is essential for faculty use as it allows for the storage and access of a large number of documents by multiple students. The desktop also features 16GB DDR5 memory, which can be upgraded to 32GB, making it well-suited for software development tasks that require high memory capacity, such as working with emulators. Additionally, it supports Wi-Fi 6, ensuring improved wireless connectivity.

Although the processor of this desktop is slightly slower compared to other alternatives like the HP Pro Tower 400 G9 PCI Desktop PC and the DELL OptiPlex Small Form Factor, the performance difference is minimal as the processors are from the same generation. When considering factors such as storage and memory, the Acer desktop provides better value in the same price range. In conclusion, the Acer Consumer Desktop | Aspire XC Series - XC-1785 offers a balanced combination of components and performance, making it an excellent choice for its value for money.

7. Monitor

 Acer Monitor - Mainstream Series - Work & Play| EK221Q H	 SAMSUNG 24" Essential Monitor S3 S31C FHD	 PRISM+ W220v
Resolution: 1920 x 1080 (Full HD) Refresh Rate: 100Hz Response Time: 1ms (VRB) Colour Gamut: 99% sRGB Panel Type: VA (Vertical Alignment) Brightness: 250 cd/m² Contrast Ratio: 3000:1 AMD FreeSync: Yes (HDMI VRR) Speakers: Yes (2 x 2W)	Resolution: 1920 x 1080 (Full HD) Refresh Rate: 75Hz Response Time: 5ms (GTG) Colour Gamut: 72% NTSC Panel Type: IPS (In-Plane Switching) Brightness: 250 cd/m² (Typical) Contrast Ratio: 1000:1 (Typical) FreeSync: Yes Speakers: No	Resolution: 1920 x 1080 (Full HD) Refresh Rate: 100Hz Response Time: 12ms (GTG) Colour Gamut: 95% sRGB Panel Type: VA (Vertical Alignment) Brightness: 220 cd/m² Contrast Ratio: 3000:1 FreeSync: Yes Speakers: No
RM 289	RM 399	RM 349

After careful comparison with two other alternatives, we decided to choose the Acer EK221QH monitor. This model stood out for offering the fastest response time of 1ms, which is particularly advantageous for web development and design, as it ensures smooth performance and responsiveness. Additionally, it covers 99% of the sRGB color gamut, delivering vibrant and accurate colors that are essential for creative work. The inclusion of AMD FreeSync technology further enhances the visual experience by reducing screen tearing and stuttering. Notably, among the three options considered, this monitor also comes at the lowest price, making it a highly cost-effective choice without compromising

on performance. Therefore, we concluded that the Acer EK221QH is the most suitable option for our needs.

8-Server:

		
PowerEdge R250 Rack Server	**HPE ProLiant DL20 Gen11 Rack Server (E-2434.16GB.1TB)**	**Lenovo ThinkSystem ST250 V2 7D8FS0Y400**
• Chassis Configuration: 3.5" Chassis with up to x4 Hot Plug Hard Drives with Backplane • Processor: Intel® Xeon® E-2324G 3.1GHz, 8M Cache, 4C/4T, Turbo (65W), 3200 MT/s • Processor Thermal Configuration: Heatsink for 80W or less CPU • Memory Capacity: 16GB UDIMM, 3200 MT/s, ECC • RAID Configuration: C3, RAID 1 for 2 HDDs or SSDs (Matching Type/Speed/Capacity) • RAID/Internal Storage Controllers: PERC H355 Adapter LP • Hard Drives: 4TB Hard Drive SAS ISE 12Gbps 7.2K 512n 3.5in Hot-Plug, AG Drive	• Processor(s): Intel® Xeon® E-2434 (4-Core, 3.4 GHz, 55W) • Memory: 16 GB UDIMM DDR5 4800 MT/s (1x 16 GB) • Hard Disk: HPE 1TB SATA 7.2K LFF LP HDD • Storage Controller: Embedded Intel® VROC SATA software RAID • Network Controller: HPE embedded 1Gb 4-port BCM5719 network adapter • Optical: HPE 9.5mm SATA DVD-ROM Optical Drive • Power Supply: (1) 290W Non-redundant Power Supply	• Processor: Intel Xeon E-2334 Processor - 3.4GHz, 8M Cache, 4 Cores • Memory: 16GB TruDDR4 3200 MHz (2Rx8, 1.2V) ECC UDIMM • Storage: 3.5" 1TB 7.2K SATA 6Gb Hot Swap 512n HDD • Optical Drive: ThinkSystem Half High SATA DVD-RW Optical Disk Drive • Hard Disk Chassis: 4x3.5" SATA/SAS Backplane • RAID Controller : ThinkSystem RAID 5350-8i PCIe 12Gb Adapter (RAID 0, 1, 10, 5, 50) • Network Controller: 2x 1GbE Onboard Ethernet ports (Broadcom BCM5720)
RM15,093.60	RM9,883.00	RM7,929.00

After evaluating the available options, we chose the HPE ProLiant DL20 Gen11 due to its well-rounded combination of performance, scalability, and cost-effectiveness. Although the Dell PowerEdge R250 offers superior specifications, its higher price makes it less practical for our needs. On the other hand, the Lenovo ThinkSystem ST250 V2, while more affordable, relies on DDR4 memory, which may limit future scalability. In contrast, the HPE ProLiant DL20 Gen11 supports DDR5 memory, ensuring better performance and future-proofing. Additionally, it features four network ports, providing redundancy and improved connectivity, and offers sufficient local storage, making it a balanced and reliable choice for our use case.

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9- Ethernet Cables:

 **cat 6a (2996AS-5G) by Videk**	 **cat 6a (21.15.2709) by Roline**	 **cat 6a (N261P-050-BL) by EATON TRIPP LITE**
• Patch Lead • RJ45 Plug to RJ45 Plug • SFTP • Double shielded for protection against EM and RF interference • Low crosstalk • 10Gbps CAT-6a standard for high demand applications • UL94V-2 Plug housing flammability	• RJ45 Plug to RJ45 Plug • UTP	• RJ45 Plug to RJ45 Plug • FUTP
RM70.1/5 m	RM190.27/20m	RM196.920/15.24m

We selected the Cat 6a (2996AS-5G) by Videk due to its high performance, affordable price, and superior shielding capabilities. This cable features double shielding (SFTP), which provides enhanced protection against both electromagnetic (EM) and radio frequency (RF) interference. Such shielding is particularly important in environments with high levels of electrical noise—like our faculty building, which contains laboratories, offices, and numerous connected devices. Given these factors, the Cat 6a (2996AS-5G) represents a reliable and cost-effective solution for our networking needs.

10- Fiber Optic Cables:

 **Fiber Optic (HFBR-RUS100Z) by BROADCOM**	 **Fiber Optic (FDPF 4001 EH) by FIBRE DATA**	 **Fiber Optic (MP007797) by MULTICOMP PRO**
• Fiber Optic Cable • Singlemode • 9µm / 125µm • 8 Fibres • 5.6 mm	• Fiber Optic Cable • Singlemode • 1mm • 1 Fibres • 2.2 mm • 65 ft • 20 m	• Fiber Optic Cable • OS2 Singlemode • 8 Fibres • 6 mm • 328 ft • 100 m
RM744.580/100m	RM133.410/20m	RM652.880/100m

We chose the Multicomp Pro OS2 Singlemode Fiber Optic Cable (MP007797) for the network backbone connections due to its combination of performance, scalability, and cost-efficiency. This cable provides 8 fibers, a length of 100 meters, and a robust 6mm outer diameter, making it suitable for demanding network environments. While the Broadcom HFBR-RUS100Z offers similar specifications, it comes at a higher cost, making the Multicomp Pro the more economical option. Additionally, compared to the Fibre Data FDPF 4001 EH, which supports only a single fiber, the Multicomp Pro allows for greater scalability and redundancy, essential for high-speed data transmission. Compliant with the OS2 standard, it supports data rates of 10Gbps, 40Gbps, and 100Gbps over long distances, while its durable construction ensures reliability during both installation and long-term use.

11-Firewalls:

 **Netgate 2100 MAX pfSense +Security Gateway**	 **N100 Firewall Router**	 **1U Rack Mount Firewall Appliance**

-The Dual core ARM v8 Cortex-A53 1.2 GHz delivers **2.20 Gbps** routing for common iPerf3 traffic and over **964 Mbps** of firewall throughput. - Flexible **5-port combination** including (1) 4-Port 1 GbE Marvell switch and a dedicated (1) GbE WAN RJ45/SFP combo port. -**8 GB eMMC Flash onboard** (or optional 128 GB M.2 SATA 2242 SSD) - **4 GB DDR4, DDR4 memory** allows more stable, higher transfer rates for memory-intensive applications (e.g. IDS/IPS). - The passive cooling system and low power draw of the Netgate 2100 provides **efficient thermal management** and **silent operation** expected from smaller devices despite its significantly higher performance.	-Powered by a **quad-core** ARM Cortex-A53 processor for efficient performance and secure routing. -Equipped with **3 Gigabit Ethernet ports**, allowing flexible WAN/LAN configurations for small networks. -Includes **2 GB DDR4 RAM and 8 GB eMMC storage** for reliable operation and advanced configurations. -Supports features like **VPN, VLANs, and advanced firewall capabilities**, ensuring strong network security. -Compact, fanless design with **low power consumption**, offering silent operation and energy efficiency.	-Powered by an **Intel Quad-Core processor** for high-performance routing and firewall operations. -Features up to **6 Gigabit Ethernet ports** for versatile and scalable network configurations. -Supports up to **8 GB DDR4 RAM,** ensuring smooth operation for resource-intensive tasks like IDS/IPS and VPN services. -Includes storage options like **SSD or HDD**, providing flexibility for logs, backups, and configurations. -Designed in a **1U rack-mountable form factor**, perfect for data centers and enterprise environments requiring compact, organized setups. -**Advanced cooling system** ensures reliable thermal
		management and consistent performance under heavy workloads.
RM 1750	RM 1500	RM 1200

We selected the 1U Rack Mount Firewall Appliance as it offers the most powerful and flexible solution, making it highly suitable for professional environments and high-budget network infrastructures. Equipped with a robust Intel Quad-Core processor and supporting up to 8 GB of memory, it is capable of efficiently managing resource-intensive tasks such as network security and private connections. The device includes six Ethernet ports and supports various storage configurations, allowing it to adapt seamlessly to complex network requirements. Its rack-mountable design ensures easy integration into professional setups, while the advanced cooling system maintains optimal performance even under heavy workloads. These features make it a reliable and scalable choice for large networks and data center applications.

12-Modems:

 **A10 Networks AX-3000-GCF Application Delivery Controllers (REF)**	 **Nighthawk®Mid/high-Split Multi-Gig 2.5Gbps Cable Modem**	 **Nighthawk® Multi-Gig 2.5Gbps Cable Modem for Xfinity Voice**
-Powered by **advanced multi-core processors**, delivering high-performance application delivery and load balancing. -Supports up to **8 Gigabit Ethernet ports**, enabling robust and scalable network integration. -Provides up to **15 Gbps of application throughput** for seamless traffic management in demanding environments. -Features **advanced Layer 4-7 traffic managemen**t	-Supports **2.5Gbps multi-gig speeds**, designed for ultra-fast internet connectivity with mid/high-split compatibility. -Certified for major ISPs, including Xfinity, Spectrum, and Cox, ensuring wide compatibility. -Features **DOCSIS 3.1 technology**, offering 32x8 channel bonding for reliable	-Supports **2.5Gbps multi-gig internet speeds**, ideal for high-performance home and office networks. -Certified for **Xfinity Voice** with support for up to two phone lines for reliable digital voice service. -Powered by **DOCSIS 3.1 technology**, offering 32x8 channel bonding for ultra-fast and stable connectivity.

capabilities for optimized application performance and security. -Includes **hardware SSL acceleration** for efficient and secure encrypted traffic handling. -Designed for **enterprise-grade reliability** with a rack-mountable 1U form factor, suitable for data centers and large-scale deployments.	high-speed performance. -Includes a **multi-gig Ethernet port**, enabling seamless integration with high-speed routers and devices. -**Compact, sleek design** with efficient cooling for stable and consistent operation. -Ideal for future-proofing home or office networks with support for high-bandwidth applications like **4K streaming and gaming**.	-Features a **multi-gig Ethernet port**, enabling seamless integration with high-speed routers and devices. -Compact, modern design with efficient cooling for **consistent operation.** -Perfect for **streaming, gaming, and voice services**, ensuring a future-proof network setup.
RM 1800	RM 1300	RM 1100

For our network project, we selected the Nighthawk® Mid/High-Split Multi-Gig 2.5Gbps Cable Modem due to its ability to deliver exceptionally fast internet speeds, which are crucial for ensuring seamless connectivity across our devices. With a maximum speed of up to 2.5Gbps, the modem is well-suited for handling high-bandwidth activities such as video streaming and online gaming, without experiencing performance degradation.

An important advantage is its compatibility with major internet service providers, which simplifies the setup process and enhances its overall reliability. Additionally, the modem features a sleek and compact design, making it ideal for space-saving installations, and its efficient cooling system helps maintain stable and consistent performance by preventing overheating.

In conclusion, the Nighthawk® modem offers the speed, reliability, and future scalability required for our project, making it the most suitable choice for our networking needs.

13- Wireless Access Point:

 **Cisco Access Point AIR-AP1832I-K-K9 (Refurbished)**	 **TP-Link EAP653 AX3000**	 **ICR-2431**
-Features **802.11ac Wave 2** technology, delivering high-speed Wi-Fi performance with data rates up to 1.3 Gbps. -Equipped with **3x3 MU-MIMO (Multi-User, Multiple Input, Multiple Output)** for efficient device connectivity and enhanced user experience. -Supports dual-band operation (2.4 GHz and 5 GHz) for flexible deployment and reduced interference. -Includes **beamforming technology** to optimize wireless signal strength and coverage. -Designed with **integrated security features**, such as WPA3 and rogue device detection, ensuring reliable and secure network access. -**Compact and durable design**, ideal for medium to large-scale indoor deployments like offices and classrooms. .	-Supports **Wi-Fi 6 (802.11ax)** technology, offering ultra-fast speeds of up to 2976 Mbps (2402 Mbps on 5 GHz and 574 Mbps on 2.4 GHz). -Features **MU-MIMO and OFDMA** to handle multiple devices efficiently, reducing latency and boosting overall network performance. -Equipped with **seamless roaming** to ensure uninterrupted connectivity as users move across the network. -Includes **Power over Ethernet (PoE+) support**, simplifying deployment and reducing cable clutter. -Offers **advanced security protocols**, including **WPA3**, to safeguard user data and enhance network protection. -**Compact, sleek design** suitable for ceiling or wall mounting, ideal for offices, schools, and hospitality environments.	-Powered by an **advanced processor**, offering reliable performance for industrial and IoT applications. Supports **dual SIM failover** and multiple WAN options (Ethernet, LTE) for uninterrupted connectivity. -Features **Gigabit Ethernet ports**, serial interfaces, and digital I/Os for versatile integration with various devices and systems. Includes robust security features like VPN, firewall, and encryption to protect data and ensure secure communication. -Designed for **rugged environments** with a compact, durable enclosure and wide operating temperature range. -Supports **remote management** and monitoring through intuitive software, ideal for industrial automation, transportation, and smart city deployments.
RM 1400	RM 580	RM 1300

We chose the TP-Link EAP653 AX3000 due to its integration of the latest Wi-Fi 6 technology, which enables high-speed performance of up to 2976 Mbps. This capability is particularly beneficial for our network environment, as it supports multiple simultaneous connections without compromising speed or stability.

The EAP653 also facilitates seamless mobility, allowing users to move freely across spaces—such as offices or classrooms—without experiencing connectivity loss. Furthermore, its support for Power over Ethernet (PoE) simplifies installation by requiring only a single cable for both power and data transmission.

In addition to performance and ease of deployment, the device includes robust security features to help protect sensitive data. Its compact and unobtrusive design makes it suitable for installation in a variety of locations.

Overall, the TP-Link EAP653 AX3000 offers an optimal balance of speed, flexibility, and security, making it the most appropriate choice for our network requirements.

14-Patch Panel:

 **TRENDnet 24-Port Cat6A Shielded Patch Panel**	 **Commscope AMP Cat6 Category 6 System 24 Port Patch Panel**	 **ICC Patch Panel, CAT 6A, 48-Port, 2 RMS (ICMPP0486B)**
-Supports **24 shielded RJ45 ports**, designed for high-performance Cat6A Ethernet connections. Provides **10 Gigabit Ethernet support**, ensuring reliable, high-speed data transmission for demanding networks. -Features a **shielded design** to reduce electromagnetic interference (EMI) and maintain signal integrity. -Includes **numbered ports** and a labeling area for easy cable identification and management. -**Durable metal housing** offers long-lasting performance and reliability in rack-mount installations. -Ideal for **professional environments** like data centers, server rooms, and structured cabling projects. .	-Features **24 RJ45 ports**, designed for efficient cable management in Cat6 Ethernet networks. -Supports **10 Gigabit Ethernet**, delivering high-speed and reliable performance for demanding applications. Includes a **color-coded wiring label** for easy installation and reduced setup errors. -Built with **durable construction** for long-lasting performance in professional environments. -Designed for **seamless integration** into standard 19-inch rack systems, ensuring a neat and organized setup. Ideal for structured cabling solutions in offices, data centers, and enterprise networks.	-Equipped with **48 RJ45 ports**, supporting high-performance Cat6A Ethernet connections. Certified for **10 Gigabit Ethernet**, ensuring reliable and fast data transmission. -Features a **2U rack-mount design**, fitting seamlessly into standard 19-inch racks for organized installations. -Includes **color-coded labels** for T568A and T568B wiring configurations, simplifying setup and maintenance. -Built with a **durable metal frame**, providing long-lasting performance in demanding environments. -Perfect for **large-scale structured cabling projects** in data centers, server rooms, and enterprise networks.
RM 460	RM 580	RM 1700

We selected the ICC Patch Panel, CAT 6A, 48-Port, 2 RMS as it offers the most suitable solution for our high-budget network project. With 48 ports, it allows for the simultaneous connection of a large number of devices, making it ideal for extensive network infrastructures. Its support for 10 Gigabit Ethernet ensures high-speed and reliable data transmission, which is essential for demanding applications.

The 2U rack-mount design facilitates easy integration into standard server racks, helping maintain a neat and organized setup. Additionally, the presence of color-coded labels simplifies the wiring process and reduces installation time by providing clear guidance.

Constructed from durable materials, this patch panel is built to withstand the demands of high-traffic environments, offering both longevity and stability. Overall, it delivers the performance, scalability, and efficiency required for the success of our network deployment.

Meeting Minutes 1

Date/Time	8 May 2025 9:00pm		
Location	Online (google meet)		
Agenda	Task distribution for task 3		
Meeting MC	Garrah Thabit Mohammed		
Attendance			
Name		Time	Reason For Absence
Garrah Thabit Mohammed		9:00 pm	-
Mohammed Abdulrahim		9:00 pm	-
LEE TUNG ERN		9:00 pm	-
MOHAMMED ALSAKKAF		9:00 pm	-
Minutes			
No.	Items Discussed	Ideas/Suggestions and Person Giving It	Person in Charge & Date

1.	The devices (network and end-user devices) that will need to achieve the objective of fulfilling the requirements and needs of the organization	Each group member contributed their input regarding the essential devices required for the organization. In total, 14 different types of devices were identified to meet the operational needs. Garrah emphasized the importance of printers, copiers, smart TVs, desktop towers, and monitors for the effective functioning of faculty activities. Mohammed highlighted the necessity of routers, switches, servers, Ethernet cables, and fiber optic cables to support network infrastructure. Meanwhile, Lee Tung pointed out that firewalls, modems, wireless access points, and patch panels are also essential for ensuring secure and reliable connectivity.	- GARRAH - MOHAMMED - LEE TUNG
2.	Task Distribution	Garrah volunteered to conduct research on printers, copiers, smart TVs, desktop towers, and monitors.	GARRAH

		Mohammed took on the task of researching routers, switches, servers, and network cabling.	MOHAMMED
		LEE TUNG volunteered to do research on firewalls, modems, wireless access points, and patch panels	LEE TUNG
3.	Record of the discussion	Recorded all the devices needed and send it to the group	GARRAH
		Documented which group member was responsible for researching each device.	ALSAKKAF
4.	Decide next discussion (meeting)	we agreed to hold a meeting on 12 May 2025 at 8:00 PM to discuss and finalize the quantity of each device required, based on both the pricing information gathered during our research and the overall budget available.	All members

5.	Meeting ended	Meeting ended at10:30 pm.	-

Meeting Minutes 2

Date/Time	12 May2025 - 8:00pm		
Location	Online (google meet)		
Agenda	Discuss about the quantity of devices base on budget		
Meeting MC	GARRAH		
Attendance			
Name		Time	Reason For Absence
Garrah Thabit Mohammed		8:00 pm	-
Mohammed Abdulrahim		8:00 pm	-
LEE TUNG ERN		8:30 pm	-
MOHAMMED ALSAKKAF		9:00 pm	-
Minutes			
No.	Items Discussed	Ideas/Suggestions and Person Giving It	Person in Charge & Date

1.	The result of researching	Everyone has shown the results of their research on the assigned devices, including descriptions and prices, in a table created by Othman. Each type of device was required to be sourced from 3 different brands for comparison.	All members
2.	Decide the amount of devices needed	After each member fills in the table, we will start deciding the quantities needed and choose the best devices based on our budget, one by one, following the table.	All members
3.	Reflections	For the reflection on Question 3, we summarized our individual opinions following a group discussion and recorded them in a shared Google Document, providing justifications for each device choice.	All members

4.		Once the discussion was complete and final decisions were made, each group member was assigned specific tasks to complete their part of the work promptly.	All members
5.	Meeting Ended	Meeting ended at 11.30pm	-

References

1A.Printer (HP OfficeJet Pro 8120 All-in-One Printer)
HP OfficeJet Pro 8120 All-in-One Printer - (405W3C) - Shop HP.com Malaysia

1B.Printer(Canon Pixma E410 All-In-One Printer)
Printing - PIXMA E410 (Green White) - Specification - Canon Malaysia

1C.Printer(Brother A4 + A3 Ink Tank MFC-J2340DW)
MFC-J2340DW – Brother Malaysia

2A.Copier(HP LaserJet M635fht Laser Multifunction Printer-Monochrome-Copier)
HP LaserJet M635 M635fht Laser Multifunction Printer-Monochrome-Copier – Network Hardwares

2B.Copier(Kyocera TASKalfa 2554ci Printer)
Kyocera TASKalfa 2554ci Printer, Up to 1200 x 1200 DPI Resolution, Up to 100 IPM Simplex/200 IPM Duplex Scan Speed, 25 PPM B&W/25 PPM Color Engine Speed | 1102YP3NL0 Buy, Best Price. Global Shipping.

2C.Copier(Brother MFC L9610CDN Color Laser All-in-One Copier)
Brother MFCL9610CDN MFC-L9610CDN Enterprise Color Laser All-in-One Pri – Network Hardwares

3A.Smart TV(PRISM+ Q75 Ultra)
PRISM+ Q75 Ultra – PRISM+ Malaysia

3B..Smart TV(Philips 7100 Series 75-inch 4K UHD LED TV)
Philips 7100 Series 75-inch 4K UHD LED 2024 Google TV - Black (75PUT7129/68) | Harvey Norman Malaysia
7100 series 4K UHD LED TV 75PUT7129/68 | Philips

3C..Smart TV(SAMSUNG 75" Crystal UHD DUE800 4K Smart TV)
Buy 75 Inch Crystal UHD DUE800 4K Tizen OS Smart TV | Samsung MY

4A. Router (ASUS RT-AX58U)
ASUS RT-AX88U Pro

4B. Router (Netgear RAX50)
Netgear RAX50

4C. Router (TP-Link Archer AX75)

TP-Link Archer AX75

5A. Switch (TP-Link TL-SG2428P)
TP-Link TL-SG2428P

5B. Switch (MikroTik CRS326-24G-2S+RM)
CRS326-24G-2S+RM

5C. Switch (Aruba Instant On 1930 48G 4SFP/SFP Switch JL685A)
Aruba Instant On 1930 48G 4SFP/SFP Switch JL685A

6A.Desktop(HP Pro Tower 400 G9 PCI Desktop PC)
HP Pro Tower 400 G9 PCI Desktop PC - (A3MK8PT) - Shop HP.com Malaysia

6B.Desktop(DELL OptiPlex Small Form Factor)
Dell OptiPlex Small Form Factor Desktop | Dell Malaysia

6C.Desktop(Acer Consumer Desktop | Aspire XC Series - XC-1785)
Acer Consumer Desktop | Aspire XC Series - XC-1785-14400W11B, Intel® Core™ i5 14400 | Acer Malaysia Official Store

7A.Monitor(Acer Monitor - Mainstream Series - Work & Play| EK221Q H)
Acer Monitor - Mainstream Series - Work & Play| EK221Q H | Acer Malaysia Official Store

7B.Monitor(SAMSUNG 24" Essential Monitor S3 S31C FHD)
24" Essential Monitor S3 S31C FHD LS24C310EAEXXS | Samsung Malaysia

7C.Monitor(PRISM+ W220v)
PRISM+ W220v - Productivity Monitor – PRISM+ Malaysia

8A. Server (PowerEdge R250 Rack Server)
PowerEdge R250 Rack Server

8B. Server (HPE ProLiant DL20 Gen11 Rack Server (E-2434.16GB.1TB)
HPE ProLiant DL20 Gen11 Rack Server (E-2434.16GB.1TB)

8C. Server (Lenovo ThinkSystem ST250 V2 7D8FS0Y400)
Lenovo ThinkSystem ST250 V2 7D8FS0Y400

9A. Ethernet Cable (cat 6a (2996AS-5G) by Videk)
cat 6a (2996AS-5G) by Videk

9B. Ethernet Cable (cat 6a (21.15.2709) by Roline)
cat 6a (21.15.2709) by Roline

9C. Ethernet Cable (cat 6a (N261P-050-BL) by EATON TRIPP LITE)
cat 6a (N261P-050-BL) by EATON TRIPP LITE

10A. Fiber Optic (HFBR-RUS100Z) by BROADCOM
Fiber Optic (HFBR-RUS100Z) by BROADCOM

10B. Fiber Optic (FDPF 4001 EH) by FIBRE DATA
Fiber Optic (FDPF 4001 EH) by FIBRE DATA

10C. Fiber Optic Cable (Fiber Optic (MP007797) by MULTICOMP PRO)
Fiber Optic Cable (Fiber Optic (MP007797) by MULTICOMP PRO)

11.A. Firewall (Netgate 2100 MAX pfSense+ Security Gateway)
https://shop.netgate.com/products/2100-max-pfsense

11.B Firewall (N100 Firewall Router)
https://www.iwillminipc.com/network-security/firewall-micro-appliance/n100-firewall-router.html?_gl=1*1pn3wle*_up*MQ..*_gs*MQ..&gclid=CjwKCAiAjKu6BhAMEiwAx4UsAo13_uF0ttIOB2nnYFXQxv9-QR2qSEOoyXhIflCh4enDHA7OFAU4QRoCnkQQAvD_BwE

11.C FireWall (1U Rack Mount Firewall Appliance)
https://www.iwillminipc.com/network-security/1u-rackmount-network-server/1u-rack-mount-firewall-appliance.html?_gl=1*1cd376p*_up*MQ..*_gs*MQ..&gclid=CjwKCAiAjKu6BhAMEiwAx4UsAjxkQThSqEGlV6TYH73fMeFmGmPweocgUl3i6GO5OFmGWSyP9XKHpBoCWtYQAvD_BwE

12.A.Modem (A10 Networks AX-3000-GCF Application Delivery Controllers (REF))
https://fishbone.com.my/a10-networks-ax-3000-gcf-ref?gad_source=1&gclid=CjwKCAiAjKu6BhAMEiwAx4UsAqjSXxQIV_JP8SWYvk259KR28_YUDf9N0mnoysFTp-trH6jq3BP-oxoC91MQAvD_BwE

12. B.Modem (Nighthawk® Mid/high-Split Multi-Gig 2.5Gbps Cable Modem)
https://www.netgear.com/home/wifi/modems/cm3000/

12.C. Modem (Nighthawk® Multi-Gig 2.5Gbps Cable Modem for Xfinity Voice)
https://www.netgear.com/home/wifi/modems/cm2050v/

13. A.Wireless Access point (Cisco Access Point AIR-AP1832I-K-K9 (Refurbished))
https://fishbone.com.my/refurbished-cisco-access-point-air-ap1832i-k-k9?srsltid=AfmBOoq08iYeOCP2jaWMvr05MRD4m0S49Ddrn6M1VHp1asHvIk0CrDsw

13.B. Wireless Access point (TP-Link EAP653 AX3000)
https://www.networkhardwares.com/en-my/products/tp-link-eap653-tp-link-ax3000-ceiling-mount-dual-band-wi-fi-6-access-point-eap653?variant=41654228123853&utm_source=google-ads&utm_campaign=&utm_agid=&utm_term=&creative=&device=c&placement=&gad_source=1&gclid=CjwKCAiAjKu6BhAMEiwAx4UsAjbkw0TEehPwnT_ac9y0k_5h7DYQCP2qcoypaUUl4JVzncVmtS9cgxoCnSAQAvD_BwE

13.C. Wireless Access point (ICR-2431)
https://www.iotmart.com/en-en/s/product/detail/01t2y000000czAQAAY?language=en_US&gad_source=1&gclid
=CjwKCAiAjKu6BhAMEiwAx4UsAn2nNoblN3h0QtIiBIq8_2bb9IUOYZ2qFZYkvjOtX9ihA1NAvcHG-xoCy2 4QAvD_BwE

14.A Patch Panel (TRENDnet 24-Port Cat6A Shielded Patch Panel)
https://www.networkhardwares.com/en-my/products/trendnet-tc-p24c6as-trendnet-24-port-cat6a-patch-panel-shielded-tc-p24c6as?variant=47557555978445&utm_source=google-ads&utm_campaign=&utm_agid=&utm_term=&creative=&device=c&placement=&gad_source=1&gclid=CjwKCAiAjKu6BhAMEiwAx4UsAlgGKtVseXP8vuOi_ia0RsiHAirNiOW-tvPiaKPOBANFSYEakx83UBoCvz8QAvD_BwE

14.B Patch Panel (Commscope AMP Cat6 Category 6 System 24 Port Patch Panel)
https://www.sysnapse.com/commscope-amp-cat6-category-6-system-24-port-patch-panel?gad_source=1
&gclid=CjwKCAiAjKu6BhAMEiwAx4UsAul9LVApdQJuW6UBuAcUvBNkfDNx5QvEFRs7890rhxmiFLbNfwtBBRoCLEEQAvD_BwE

14.C Patch Panel (ICC Patch Panel, CAT 6A, 48-Port, 2 RMS (ICMPP0486B))
https://www.networkhardwares.com/en-my/products/icc-icmpp0486b-icc-patch-panel-cat-6a-48-port-2-rms-icmpp0486b?variant=47992374493389&utm_source=google-ads&utm_campaign=&utm_agid=&utm_term=&creative=&device=c&placement=&gad_source=1&gclid=CjwKCAiAjKu6BhAMEiwAx4UsAuB8rcGXtePc0yR3BiIawRtLQ3frq1hocXcRuQuDc8qAwgJ0dyaPcRoCluAQAvD_BwE



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

SCSR2213 NETWORK COMMUNICATIONS

Task#4

Making The Connections - LAN And WAN

Group 4: Linux

SECTION: 5

Lecturer: Dr. Adnan Khalid

Name	Matric Number
Garrah Thabit Mohammed	A24CS4013
Mohammed Abdulrahim	A24CS4026
LEE TUNG ERN	A24CS0103
MOHAMMED ALSAKKAF	A23CS4026

SCSR2213

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- ● Floor Diagram.
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SCSR2213



In the earlier tasks, we laid a strong foundation for the project. In Task 1, our team, "Linux," was formed, and we developed a comprehensive floor plan for the new two-story Faculty of Computing building, along with documenting our preliminary discussions in meeting minutes.

Task 2 involved a thorough analysis where we formulated over 10 focused questions to better understand the requirements and gather insights. Through detailed research and consultations, we assessed the current and future needs of the project, evaluated its feasibility, and made necessary revisions to our design to align with the expectations of stakeholders.

For Task 3, we selected the most suitable LAN devices that meet the objectives of the case study, ensuring that the building would be equipped with a robust and secure network infrastructure capable of meeting both current and future demands.

In this task, we will focus on designing the physical network infrastructure, including strategic placement of devices, cabling, and inter-floor connections to seamlessly integrate the new rooms into the overall network design. We will carefully plan for both horizontal and vertical cabling, choosing the right media (twisted pair, fiber optic, or wireless) based on the specific requirements. The ultimate goal is to create a reliable, high-performance network infrastructure for the new building.

SCSR2213

- **Working areas: First Floor Working Areas**

The ground floor of the building, as shown in Figure 1, includes six distinct work areas.

Work Area 1 is the Server Room, which is essential for the IT infrastructure. It houses five server racks responsible for managing data, applications, and network performance throughout the building. In addition, there are two workstations in the room, allowing IT staff to monitor and manage server operations effectively. These devices are critical to ensuring the smooth operation of the facility and supporting the overall network.

Work Area 2 is the Hybrid Classroom, which is equipped with two large TVs for displaying teaching materials and presentations, making it easier for instructors to share content with students. The classroom also contains a workstation for the instructor to facilitate lesson delivery and manage activities, ensuring a better learning experience.

Work Area 3 is the Video Conference Room, dedicated to virtual meetings and collaborations. This room features a large TV for video calls, presentations, and screen sharing, enabling participants to interact seamlessly.

Work Area 4 is the Lecturer's Office, consisting of 18 individual rooms. Each room includes a workstation, providing lecturers with a private, quiet space to carry out administrative and academic work.



Figure 1: Work Areas in ground floor

Work Area 5 is the Academic Office, a space designed for both individual and collaborative work. It is equipped with 27 workstations to accommodate staff and students. Additionally, there are three TVs for presentations and information display. To support document management, two printers and two photocopiers are available.

Work Area 6 is the Meeting Room, which has a large TV for content display, facilitating meetings and enabling participants to share materials and collaborate effectively.

SCRIBD

This setup ensures that each work area is optimally designed to meet the needs of different functions within the building.

Figure 2 illustrates the work areas located on the first floor, which consists of five main workspaces.

Work Area 7 is the Student Lounge, designed as a space for students to relax, socialize, and collaborate. It features comfortable sofas, a large meeting table surrounded by chairs, and small tables, offering a flexible environment suitable for both individual and group activities. The lounge is also equipped with a TV for informational purposes and a photocopier to assist students with printing and copying tasks.

Work Area 8 is the Cisco Network Lab, a specialized facility for networking studies and hands-on applications. This lab is furnished with 31 workstations, including 30 for students and 1 for the lecturer. Additionally, there is a small storage room with three cabinets to organize tools, equipment, and other materials essential for networking activities, ensuring the lab remains organized and functional.

Work Areas 9 and 11 are the General Labs, each of which is equipped with 31 workstations: 30 for students and 1 for the lecturer. Both labs also feature a small storage area with three cabinets to store tools, equipment, and materials needed for various lab activities.

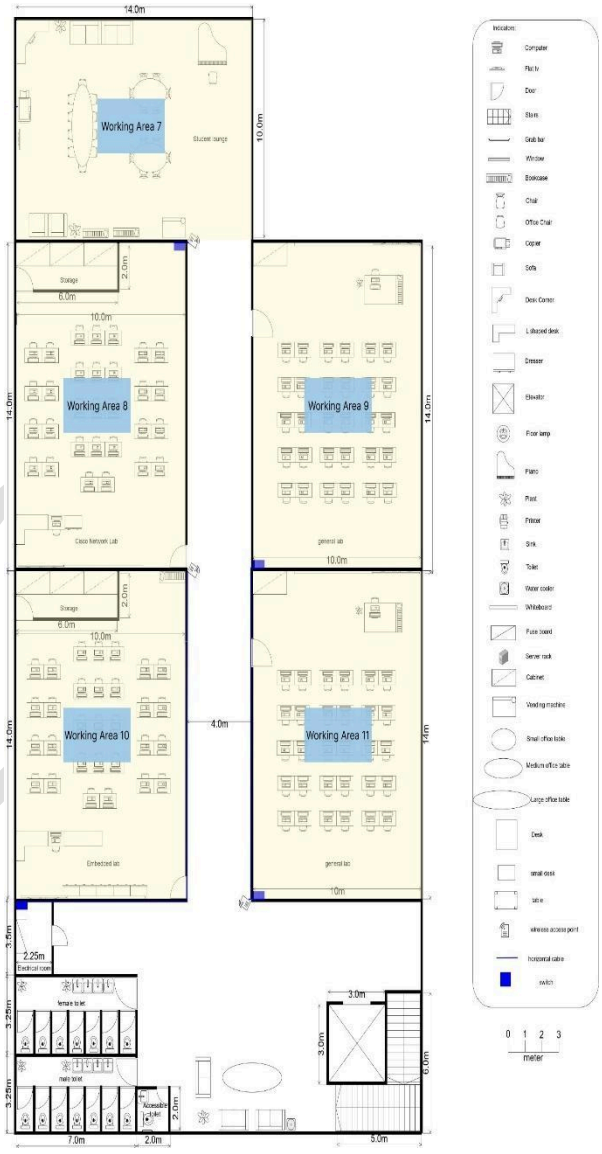


Figure 2: Work Areas in first floor

Work Area 10 is the Embedded Lab, dedicated to hardware programming and embedded systems projects. It includes 31 workstations, 30 for students and 1 for the lecturer. Like the other labs, the Embedded Lab also has a small storage space with three cabinets to store tools, components, and other necessary resources.

In summary, Figures 4.1.1 and 4.1.2 showcase a total of 11 work areas across the building, each carefully designed to support the diverse functions and activities within the facility.

The distribution and connection of devices

3.1 Network Distribution Diagram:

This diagram illustrates the network connections between the devices in the faculty building. It mainly highlights the wired and wireless links that connect the server room to the other rooms across the building.

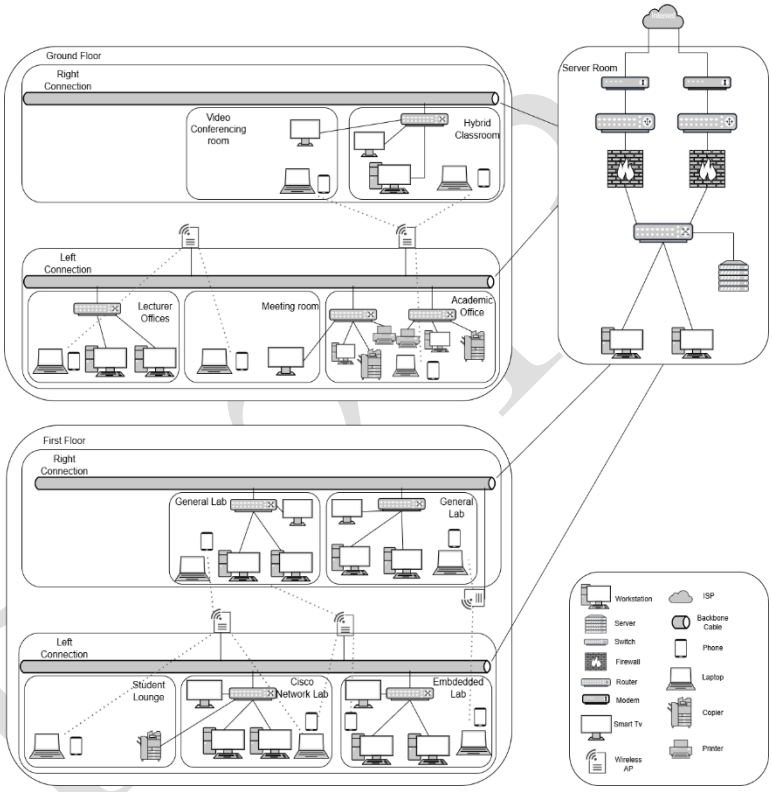


Figure 3.1.1 Network Distribution Diagram

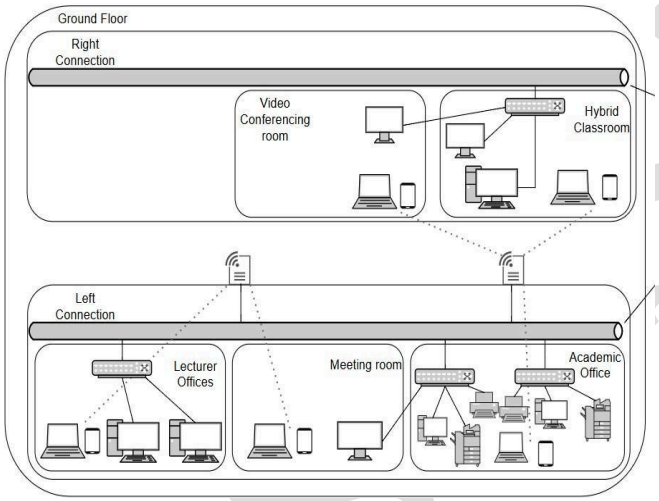


Figure 3.1.1.1 Network Distribution Diagram - Ground Floor

Ground Floor:

As shown in Figure 3.1.1.1, there are two primary connections from the server room: one on the right and one on the left. The right connection includes a single switch located in the Hybrid Classroom, which then extends to the Video Conference Room. On the left, there are three switches: two in the Academic Office to support the heavy workload, and one in the Electrical Room that connects to the lecturers' offices. Additionally, the ground floor is equipped with two wireless access points to ensure wireless internet access throughout the area.

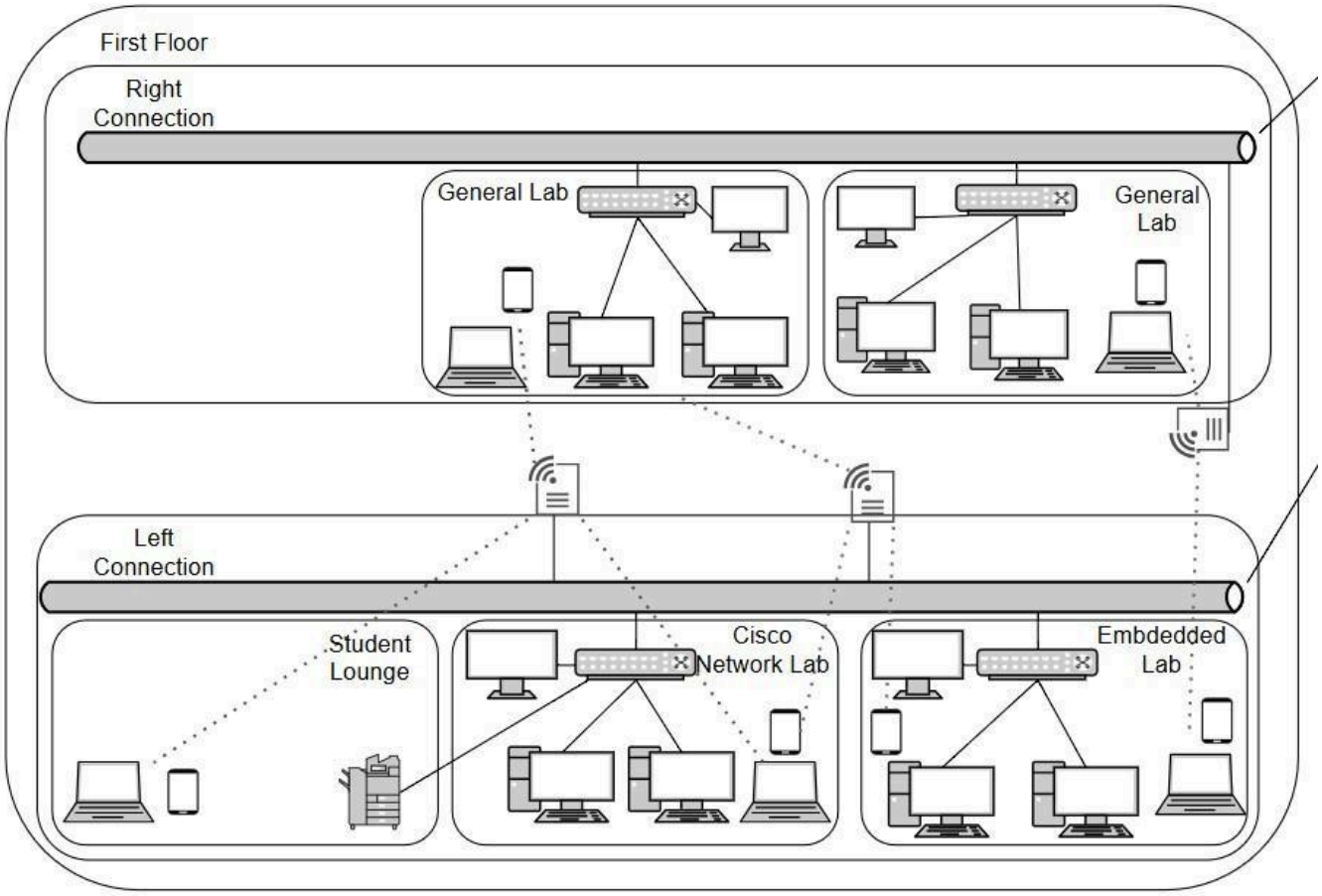


Figure 3.1.1.2 Network Distribution Diagram - First Floor

First Floor:

The first floor follows a similar setup to the ground floor, as shown in Figure 3.1.1.2, with both right and left connections. The right connection is linked to two switches, one in each General Lab, and includes a wireless access point. On the left side, there are also two switches: one in the Embedded Lab and one in the Cisco Network Lab. Additionally, the first floor is equipped with two wireless access points to ensure complete wireless coverage throughout the area.

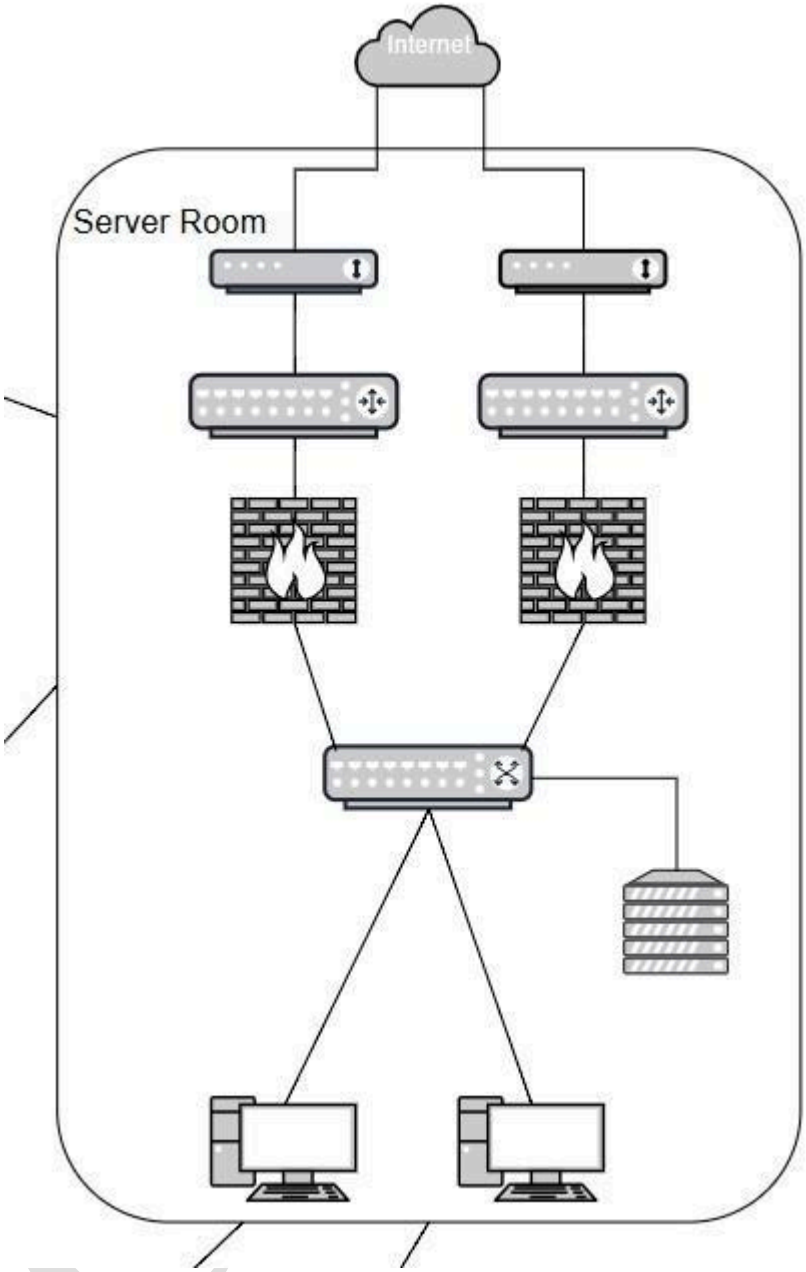


Figure 3.1.1.3 Network Distribution Diagram - Server Room

Server Room:

As seen in Figure 3.1.1.3, the server room utilizes a 2-tier network architecture, which allows it to manage the network traffic of the faculty building efficiently. This architecture is also more cost-effective compared to a 3-tier setup. The server room is equipped with an additional modem, router, and firewall to ensure redundancy, making the network more resilient. If one router fails, the other can continue managing the traffic until the problem is resolved. Furthermore, the server room contains two workstations that are directly connected to the core switch, allowing for better control over the network. The server itself is also linked to the core switch to ensure a stable and reliable connection.

- **Room Level Network Distribution Diagram:**

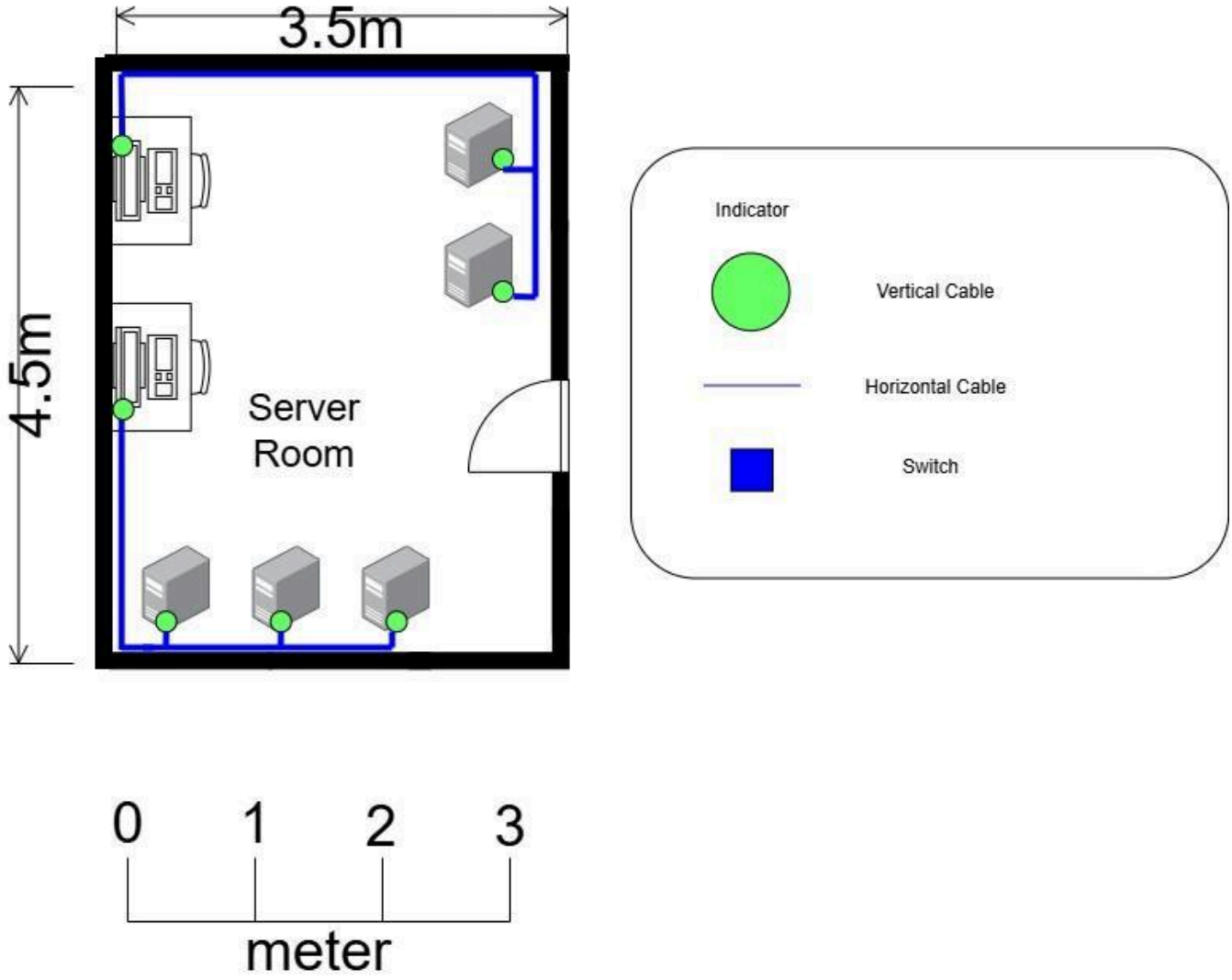


Figure 3.2.1 : Server room

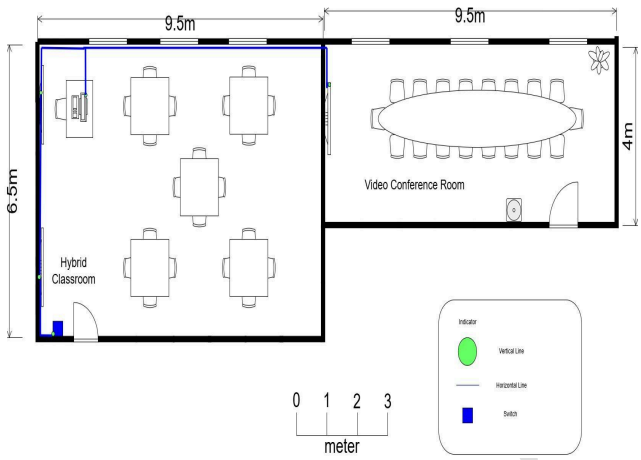


Figure 3.2.2 : Hybrid Classroom & video conference room

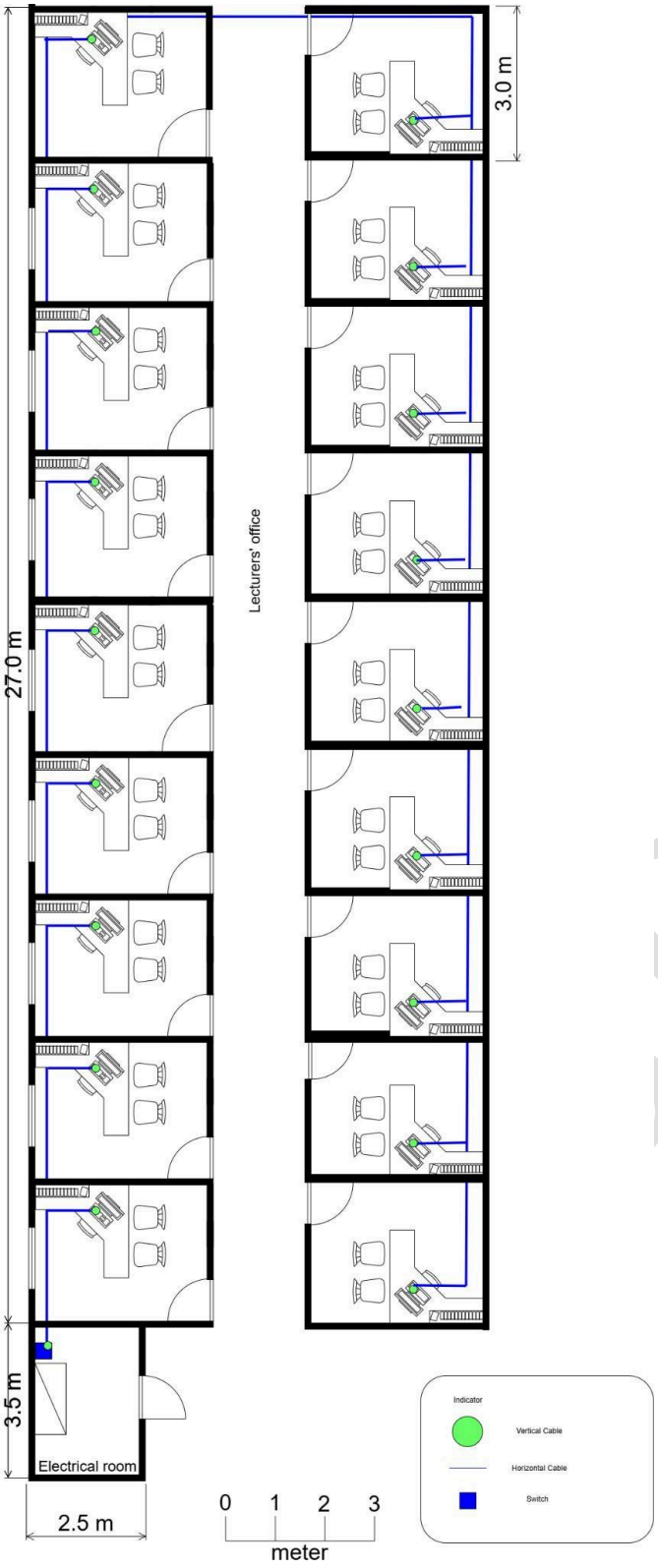


Figure 3.2.3 : Lecturer's Office & Electrical Room

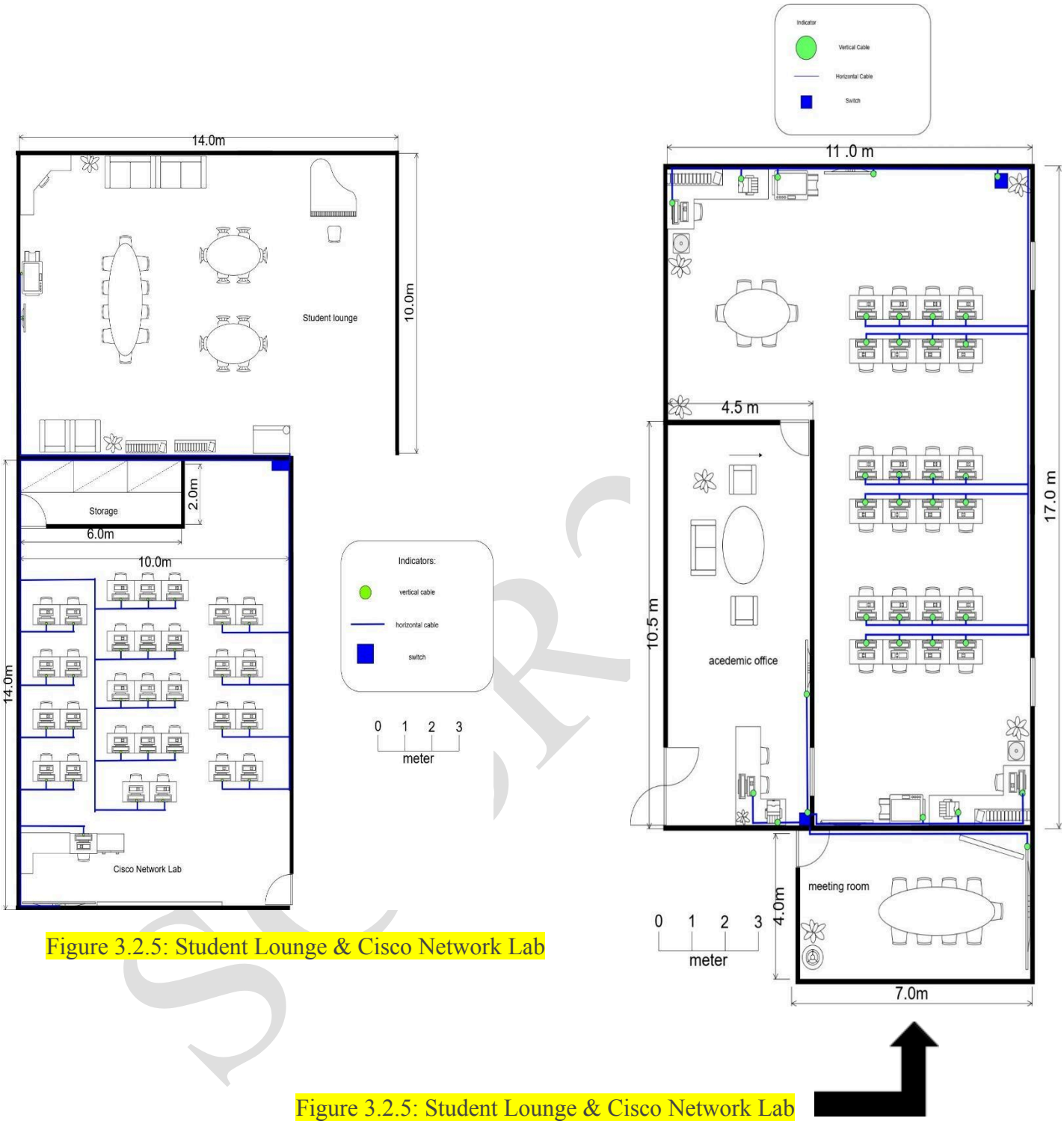
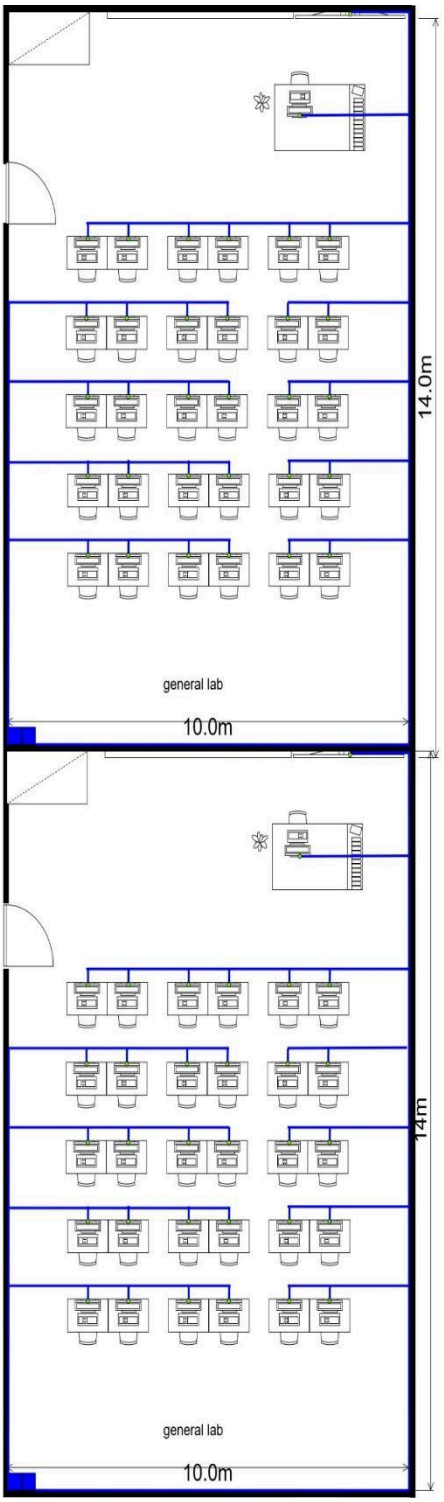
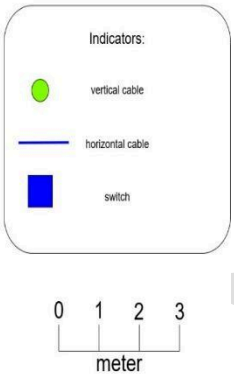
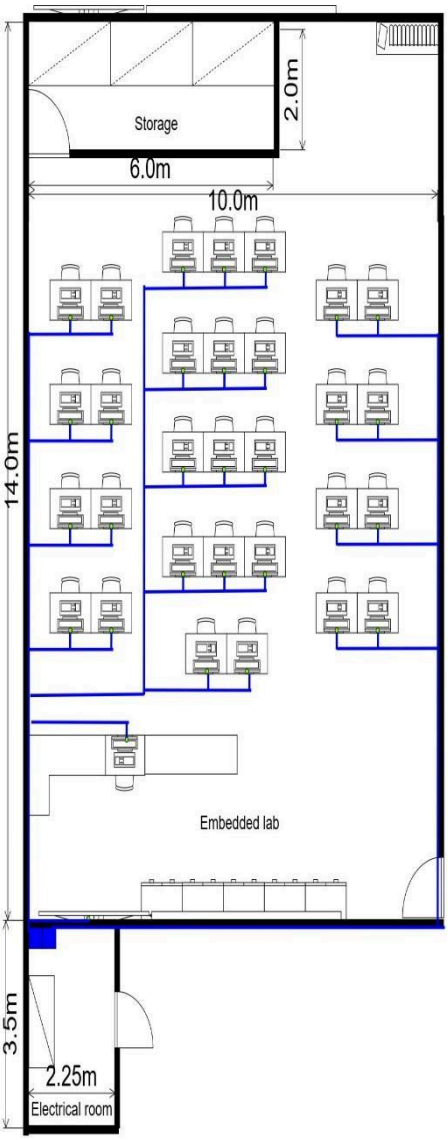


Figure 3.2.5: Student Lounge & Cisco Network Lab

Figure 3.2.5: Student Lounge & Cisco Network Lab







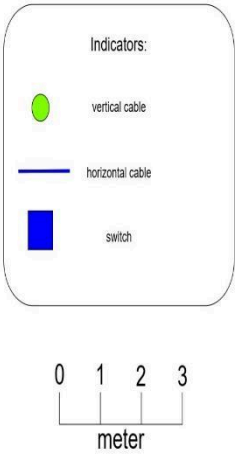


Figure 3.2.6 : Embedded Lab & Electrical room

Figure 3.2.7 : General Labs

Floor Diagram:

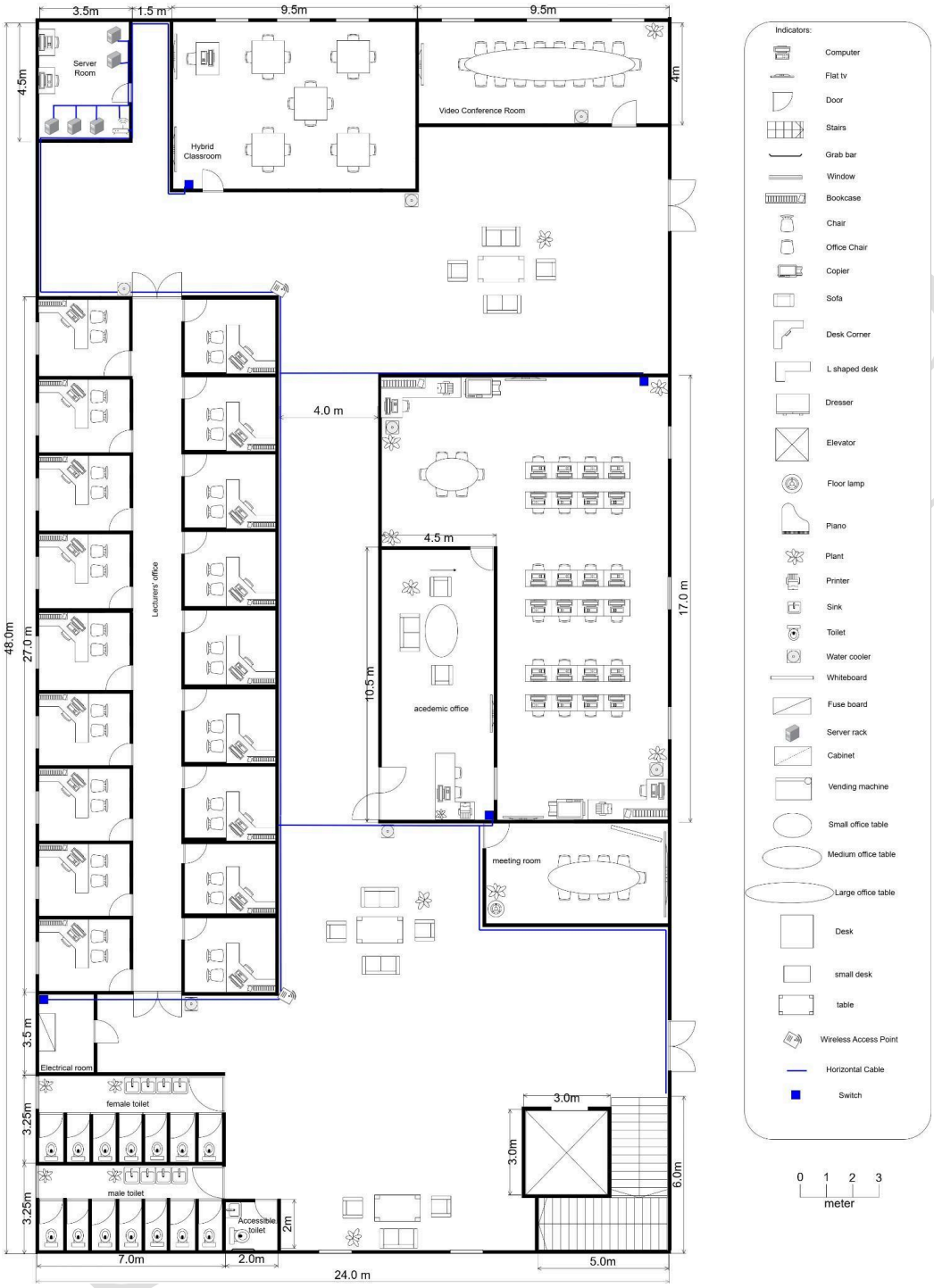


Figure 3.3.1 : Ground Floor

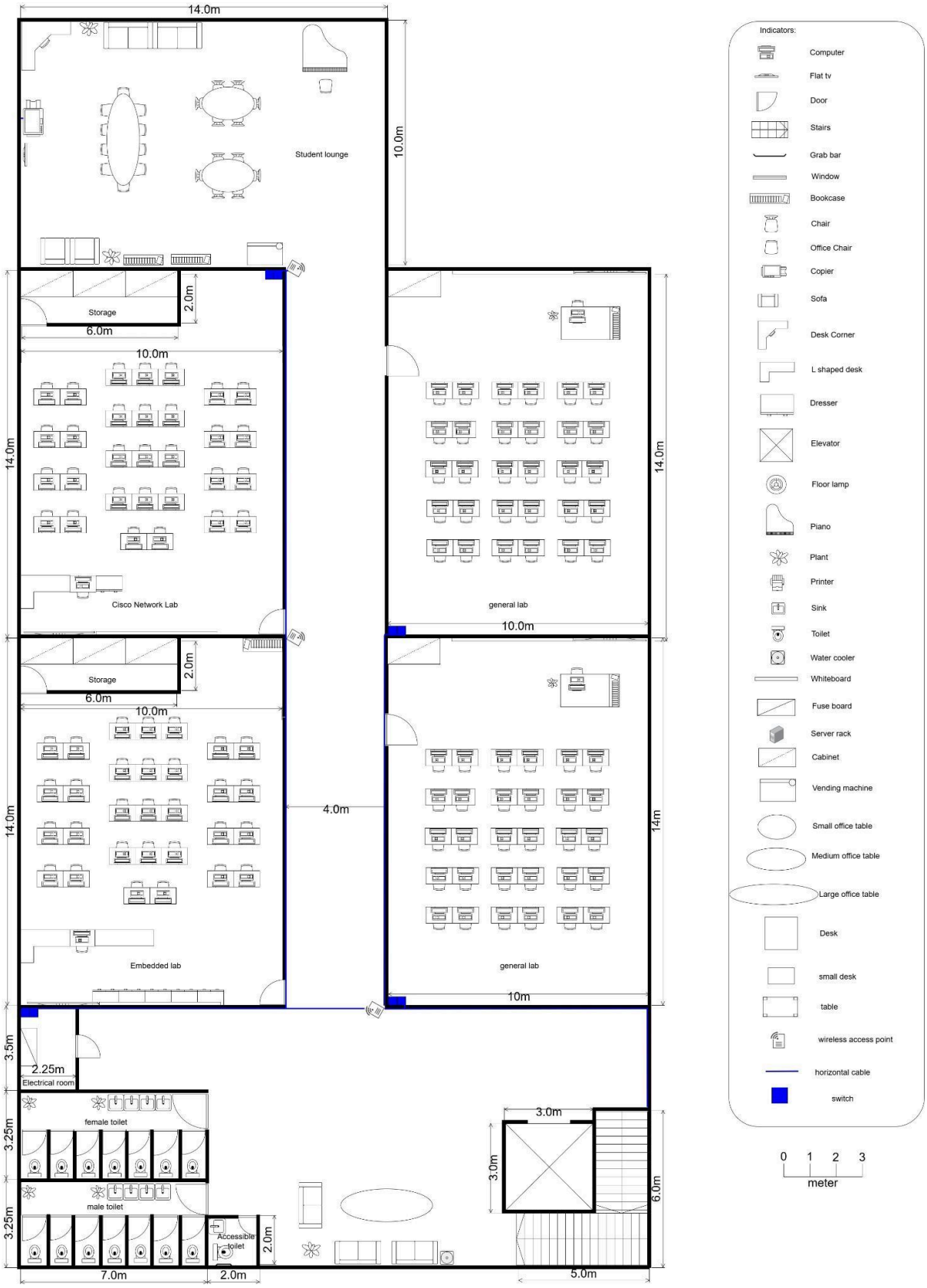

14.0m
Student lounge
10.0m
Storage
2.0m
6.0m
10.0m
14.0m
Cisco Network Lab
general lab
14.0m
10.0m
Storage
2.0m
6.0m
10.0m
4.0m
14.0m
Embedded lab
general lab
14m
10m
3.5m
2.25m
Electrical room
female toilet
3.25m
male toilet
3.25m
Accessible toilet
2.0m
7.0m
2.0m
3.0m
3.0m
6.0m
5.0m
Indicators:
Computer
Flat tv
Door
Stairs
Grab bar
Window
Bookcase
Chair
Office Chair
Copier
Sofa
Desk Corner
L shaped desk
Dresser
Elevator
Floor lamp
Piano
Plant
Printer
Sink
Toilet
Water cooler
Whiteboard
Fuse board
Server rack
Cabinet
Vending machine
Small office table
Medium office table
Large office table
Desk
small desk
table
wireless access point
horizontal cable
switch
0 1 2 3
meter

Figure 3.3.2 : First Floor

Number of connections, patch cords and switch ports needed:

- **Number of Switch Ports Needed:**

Area	Number of Switch Ports Needed
Hybrid Classroom	3
Video Conference Room	1
Academic Office	34
Meeting Room	1
Lecturer's Office	18
Cisco Network Lab	32
Embedded Lab	32
General Lab 1	32
General Lab 2	32
Student Lounge	2
Total	187

- **Number of Patch Cords Needed:**

Area	Number of Patch Cords Needed
Hybrid Classroom	3
Video Conference Room	1
Academic Office	34
Meeting Room	1
Lecturer's Office	18
Cisco Network Lab	32
Embedded Lab	32
General Lab 1	32
General Lab 2	32

Student Lounge	2
Total	187

- **Number of Connectors Needed:**

Area	**Number of Connectors Needed**
Server Room	14
Hybrid Classroom	6
Video Conference Room	2
Academic Office	68
Meeting Room	2
Lecturer's Office	36
Cisco Network Lab	64
Embedded Lab	64
General Lab 1	64
General Lab 2	64
Student Lounge	4
Total	374

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Identifying cable lengths And Prices:

- **Lengths and Prices Table:**

Area	Length (m)		Total Length (m)	Total Price of Cable Used (RM)
	Horizontal	Vertical		
Ground Floor				
Room				
Server Room	9.7	2.8	12.5	83.38
Hybrid Classroom	17.6	1.6	19.2	269.18
Video Conference Room	1.4	0.2	1.6	22.43
Academic Office	126.9	73.4	200.3	2808.21
Meeting Room	7.4	0.6	8	112.16
Lecturer's Office	79.2	22.6	101.8	1427.24
Ground Floor Room Total			343.4	4722.6
First Floor				
Room				
Cisco Network Lab	84.9	25	109.9	1540.80
Embedded Lab	81.2	25	106.2	1488.92
General Lab 1	92.25	25	117.25	1643.85
General Lab 2	92.25	25	117.25	1643.85
Student Lounge	16	0.6	16.6	232.73
First Floor Room Total			467.2	6550.15
Path			Length(m)	Price of cables used (RM)

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Total Path Length of Ground Floor	110.1	1543.60
Total Path Length of First Floor	69.6	975.79
Total (Ground Floor Rooms + Fi		13792.14 RM



In the previous section, we discussed how devices in the building are interconnected and how the network is distributed. To ensure that these connections are reliable, selecting the right type of cables is crucial. After discussing as a team, we decided to use a combination of CAT6 twisted-pair cables and fiber optic cables.

CAT6 twisted-pair cables will be used for the internal wiring within the building, such as in labs and lecture halls. These cables are ideal because they support high data transfer speeds of up to 10 Gbps. The thicker copper wires in CAT6 cables help reduce heat, making them capable of handling high power levels. This not only saves time during installation but also reduces costs. We opted for the Double Shielded Twisted Pair (STP) version of CAT6, as it is more durable and has a longer lifespan.

Fiber optic cables will be used to connect the routers in the server room and link the building to the Internet Service Provider (ISP). These cables transfer data using light, making them faster and more reliable. They can carry large amounts of data over long distances without any loss in signal quality. Fiber optic cables are essential for ensuring a stable and strong connection.

In conclusion, the building will require both CAT6 and fiber optic cables. This combination will provide fast, reliable connections within the building and ensure a strong link to the ISP.

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Meeting Minutes 1

Date/Time	7June 2025 10:00pm
Location	Online “google meet”
Agenda	1. The task. 2. The suggested idea. 3. Task division. 4. Working on the project.
Meeting MC	Garrah Thabit Mohammed

Attendance

Name	Time	Reason For Absence
Garrah Thabit Mohammed	10:00 pm	-
MOHAMMED ALSAKKAF	10:00 pm	-
LEE TUNG ERN	10:30 pm	-
Mohammed Abdulrahim	10:10 pm	

Minutes

No.	Items Discussed	Ideas/Suggestions and Person Giving It	Person in Charge & Date

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1.	Identifying Work Area	Garrah initially identified the work areas and discussed them with the other team members. Once everyone agreed, he mentioned that he would finalize the labels after the meeting.	Garrah
2.	Connections	discussed the design of the network distribution diagram for the floor plan, and volunteered to create the diagram later and update the connection design.	Garrah
		The group also discussed how to approach the floor plan connection design, deciding that two members would each handle the design for one floor. They would also calculate the length of cables and their costs.	The Team
		volunteered to design the network connections for the ground floor, which includes six work areas.	LEE TUNG
		Garrah volunteered to design the connection for the first floor, including 5 work areas.	Garrah

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3.	Google docs	A collaborative document was created, and a table was set up to track the required cable lengths and types for the project.	Mohammed Abdulrahim
4.	Meeting ended	Meeting ended at11:30 pm.	-

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Faculty of Computing

SCSR2213 NETWORK COMMUNICATIONS

Task#4

Group 4: Linux

SECTION: 5

Lecturer: Dr. Adnan Khalid

Name	Matric Number
Garrah Thabit Mohammed	A24CS4013
Mohammed Abdulrahim	A24CS4026
LEE TUNG ERN	A24CS0103
MOHAMMED ALSAKKAF	A23CS4026

SCSR2213

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- Brief explanation for the new task.
- Summary from the last task

2. Identifying the Network address 4

- Network Subnetting
- IP Assignation to working areas.

5. Meeting minutes 10

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In the previous tasks of our project, we have established a solid foundation for the development of the new Faculty of Computing building. In Task 1, we formed our team, "Linux" and collaborated to design a comprehensive floor plan for the two-story building. The layout includes classrooms, laboratories, and common areas, ensuring that it accommodates the needs of both students and faculty. Additionally, we documented our initial discussions in meeting minutes, which helped us track our progress and decisions.

In Task 2, we carried out a preliminary analysis to gather relevant information about the project. We formulated over 10 focused questions to clarify our requirements and gain deeper insights. Through research and consultations with stakeholders, we identified both the current and potential future needs of the building. This allowed us to assess the feasibility of the project and refine our design plan to meet the expectations of all involved parties.

Task 3 focused on selecting the appropriate Local Area Network (LAN) devices that would meet the objectives outlined in our case study. Our main goal was to ensure that the new building would have an efficient and secure network infrastructure to support both current and future demands. We evaluated various options to determine the most suitable devices for our needs.

In Task 4, we designed the physical network infrastructure of the building. This task involved deciding where each network device would be located, identifying the best cabling routes, and establishing connections between floors. We selected appropriate media, including twisted pair cables, fiber optic cables, and wireless technology, to ensure the network's reliability and efficiency. This careful planning is essential to guarantee smooth communication across all devices within the building.

Today, in Task 5, we will focus on subnetting. This task involves assigning unique IP addresses to each device in the different rooms of the building to avoid any conflicts. Using the provided network address, we will divide the larger network into smaller, more manageable subnets tailored to each lab and room. This structured approach will help us use IP addresses efficiently and ensure that the network is well-organized and free of overlaps.

These tasks have progressively built upon one another, taking us through each crucial stage of the project. With each completed task, we have gained a clearer understanding of the requirements and challenges, bringing us closer to successfully completing the Faculty of Computing building.

Identifying the Network address:

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1-Network Subnetting:

Based on the provided network address of 192.17.0.0/8, we decided to divide the subnet according to the number of devices required for each working area. The subnetting was carried out as follows:

Subnet No	Area	CIDR Subnet	IP Range	No of Usable Addresses
0	Server Room & Hybrid class room and video conference room	192.17.1.0/27	192.17.1.0 - 192.17.1.31	30
1	Lecturer's office	192.17.1.32/27	192.17.1.32 - 192.17.1.63	30
2	academic office & meeting room	192.17.1.64/26	192.17.1.64 - 192.17.1.127	62
3	General Lab 1 & 2	192.17.1.128/25	192.17.1.128 – 192.17.1.255	128
4	Cisco Network Lab & Embedded Lab and student lounge	192.17.2.0/25	192.17.2.0 – 192.17.2.127	128

IP Assignation to working areas:

Ground Floor

1. Server Room

Devices	IP Address
Default Gateway	192.17.1.1
Primary Router	192.17.1.2
Backup Router	192.17.1.3
Primary Firewall	192.17.1.4
Backup Firewall	192.17.1.5
Core Switch	192.17.1.6
SV1	192.17.1.7
SV2	192.17.1.8
SV3	192.17.1.9
SV4	192.17.1.10
PC1	192.17.1.11
PC2	192.17.1.12

2. Hybrid Classroom

Devices	IP Address
Default Gateway (same as server room)	192.17.1.1
PC3	192.17.1.13
Smart TV1	192.17.1.14
Smart TV2	192.17.1.15

CSE213

3. Video Conferencing Room

Devices	IP Address
Default Gateway (same as server room)	192.17.1.1
Smart TV3	192.17.1.16

4. Lecturer's office

Devices	IP Address
Default Gateway	192.17.1.33
PC4	192.17.1.34
PC5	192.17.1.35
PC6	192.17.1.36
…	…
PC21	192.17.1.51
WAP1	192.17.1.52

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5. Academic Office

Devices	IP Address
Default Gateway	192.17.1.65
PC22	192.17.1.66
PC23	192.17.1.67
PC24	192.17.1.68
…	…
PC48	192.17.1.92
Printer1	192.17.1.93
Printer2	192.17.1.94
Printer3	192.17.1.95
Copier1	192.17.1.96
Copier2	192.17.1.97
Smart TV4	192.17.1.98
Smart TV5	192.17.1.99
WAP2	192.17.1.100

6. **Meeting Room**

Devices	IP Address
Default Gateway (same as the Academic Office)	192.17.1.65
Smart TV6	192.17.1.101

First Floor:

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7. **General Lab 1**

Devices	IP Address
Default Gateway	192.17.1.129
PC49	192.17.1.130
PC50	192.17.1.131
PC51	192.17.1.132
…	…
PC79	192.17.1.160
Smart TV7	192.17.1.161
Switch 1	192.17.1.162
Switch 2	192.17.1.163

8. **General Lab 2**

Devices	IP Address
Default Gateway (same as general lab 1)	192.17.1.129
PC80	192.17.1.164
PC81	192.17.1.165
PC82	192.17.1.166
…	…
PC110	192.17.1.194
Smart TV8	192.17.1.195
Switch 3	192.17.1.196
Switch 4	192.17.1.197

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9. Cisco Network Lab

Devices	IP Address
Default Gateway	192.17.2.1
PC111	192.17.2.2
PC112	192.17.2.3
PC113	192.17.2.4
…	…
PC141	192.17.2.32
Smart TV9	192.17.2.33
Switch 5	192.17.2.34
Switch 6	192.17.2.35

10. Embedded Lab

Devices	IP Address
Default Gateway (same as cisco network lab)	192.17.2.1
PC142	192.17.2.36
PC143	192.17.2.37
PC144	192.17.2.38
…	…
PC172	192.17.2.66
Smart TV10	192.17.2.67
Switch 7	192.17.2.68
Switch 8	192.17.2.69

11. Student lounge

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Devices	IP Address
Default Gateway (same as cisco network lab)	192.17.2.1
Smart TV 11	192.17.2.70
Copier 3	192.17.2.71

3-Meeting Minutes:

Meeting Minutes

Date/Time	18 Jun 2025 10:00 pm		
Location	Online (google meet)		
Agenda	Distribution of IP address		
Meeting MC	Garrah Thabit Mohammed		
Attendance			
Name		Time	Reason For Absence
Garrah Thabit Mohammed		10:00 pm	-
MOHAMMED ALSAKKAF		10:00 pm	-
LEE TUNG ERN		10:15 pm	-
Mohammed Abdulrahim		10:10 pm	-
Minutes			
No.	Items Discussed	Ideas/Suggestions and Person Giving It	Person in Charge & Date

SCSR2213

1.	Network subnetting	Garrah completed the network subnetting and presented it to the team. After reviewing it, everyone agreed that it was well done and accepted the proposed solution.	Garrah
2.	The Ip address of the devices in work areas	ALSAKKAF proposed creating a table to list all the IP addresses of the devices.	ALSAKKAF
		took the responsibility of assigning the IP addresses for the work areas on the ground floor.	The Team
		took the responsibility of assigning the IP addresses for the work areas on the first floor.	The Team
3.	Meeting ended	Meeting ended at 11.30 pm.	-

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